



# Maintaining cost and ruin probability

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## Abstract

Specialized funds such as charitable trusts do not attach much value to consumption, instead, they pursue to maintain a satisfactory level of spending and avoid ruin to achieve their managerial goals. We employ an objective function tailored for studying ruin probability of a specialized fund, which implies simple analytical conditions to judge if the fund can be operable permanently. We analytically show that even if the fund has fixed portfolio weights and faces both fixed and proportional maintaining cost, there may still exist a positive probability for the fund to maintain operability permanently. Since if the stock is profitable enough, the wealth process has a large positive drift to offset effects of the fixed cost and downside risk. We extend the benchmark model to a case allowing portfolio rebalance between risky assets, and also obtain analytical expressions for optimal portfolio choices and ruin probability. Allowing portfolio selection potentially improves survival probability. Finally, we provide conditions needed to enjoy a positive probability of permanent survival when the fund can invest in a short bond (potentially with a risky asset) with stochastic nominal riskless interest rate.

**Keywords** Maintaining cost · Ruin probability · Fixed cost · Charitable trust · Permanent operability

**JEL Codes** C02 · C61 · G11 · G23

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## 1 Introduction

Many studies in the literature have proposed investment models with the objective to maximize expected utility, e.g., Merton (1971), Bodie et al. (1992), Dybvig (1995), Elie and Touzi (2006), Lim and Shin (2011), Qin (2013), Christensen and Qin (2014), Semenov (2017), Du et al. (2020) and Kwabi and Boateng (2020). Kolm et al. (2014) and Harel et al. (2018) provide a comprehensive review of the utility theories with emphasis on applications of expected utility maximization. However, some funds, e.g., charitable trusts or funds, pursue to maintain operability to achieve their managerial goals by maintaining a minimal but acceptable level of spending for an infinite period of time. As such, charitable funds might not attach much value to expenses but focus instead on maintaining operability and minimizing ruin probability. This paper employs an objective function that is tailored for studying the ruin probability of these specialized funds.

Meanwhile, the maintaining costs and the spending stipulated by rules are considerable factors that deplete the wealth of specialized fund. Usually, the maintaining costs and spending including administration fees, operating costs and philanthropic spending can be as substantial as 7% of the annual wealth of the fund. The costs consist two parts: fixed cost and variable cost. The fixed cost is needed to maintain the fund regardless of the amount of fund wealth, and the variable cost is proportional to the fund wealth. For example, Section 4942 of the Internal Revenue Code (USA) requires private foundations to distribute 5% of the fair market value of their assets each year; otherwise, the fund will be penalized with a high tax. This type of required spending is classified as proportional cost. Both fixed cost and proportional cost are the minimal spending to keep the fund operational. Managers and donors alike care about how the wealth of the fund evolves and how factors such as investment opportunities and maintaining costs would determine the ruin probability of the fund. In this paper, we construct a parsimonious workhorse model to address the following questions: How do the expected return and the return volatility affect the chances of survival of the fund? What is the ruin probability of the fund under constant proportional cost and fixed cost?

To address these questions, we first define that the trust or fund is terminated when its wealth reaches zero. By employing the martingale approach pioneered by Fleming and Rishel (1975), we establish simple models with analytical solutions. We show in a benchmark model that even if the fund has fixed target relative portfolios, there may still exist a positive probability for the fund to maintain operability permanently. Since as long as the drift of wealth is large enough to offset the effect of the proportional cost, fixed cost and the downside risk, the wealth tends to increase to infinity with a positive probability. Our model provides a neat condition to measure if the investment opportunity is profitable enough for the fund to avoid ruin at any finite time: if the ratio of the cost-adjusted expected return to return volatility is larger than half, the investment would enable the fund to maintain operability permanently.

We extend the benchmark model to a case allowing portfolio rebalancing between assets. In addition to a traditional risky asset (asset *A*), there is another asset (asset *B*) that carries another shock which is compensated by an additional expected return. In the generalized model, we can still obtain analytical expressions for both optimal portfolio choices and ruin probability. Comparing the conditions for existence of positive survival probability with that in the benchmark model, we find that portfolio selection potentially improves the survival probability. Particularly, when the expected return of asset *B* is low enough, a short position in asset *B* is able to substantially reduce the probability of ever termination

of the fund. Moreover, intuitively, the optimal portfolio weight in asset  $B$  decreases with the level of additional return volatility. The portfolio weight in asset  $B$  increases in the additional expected return, since a larger additional expected return leads to a higher total expected return and larger Sharpe ratio. Interestingly, the optimal portfolio weight in asset  $B$  is a concave function of the additional expected return. Because as the additional expected return increases, the ruin probability becomes lower and closer to zero, thus there is less room for improvement in survival probability by increasing the portfolio weight in asset  $B$ , given the return volatility unchanged. Therefore, the marginal optimal portfolio weight in asset  $B$  decreases with the additional expected return. Besides, the optimal portfolio weight in asset  $B$  decreases in the level of wealth. The intuition is that when the wealth level is low, the fund faces large ruin probability due to the existence of fixed maintaining cost. Hence, the managers have a stronger incentive to gamble and earn high risk premium from asset  $B$ , which yields extra expected return by bearing additional risk. Furthermore, we also find that the optimal portfolio weight in asset  $B$  increases with the fixed maintaining cost and the return volatility of asset  $A$ .

Since the nominal interest rate varies substantially due to the inflation fluctuations in the real world, we also investigate if the fund can maintain operability forever by investing in short bonds (potentially with a risky asset) in an environment with stochastic nominal riskless interest rate. By applying the results we have obtained in the two previous models, we provide the conditions of positive survival probability for both cases: (1) investors can only invest in a short bond and (2) investors can invest in both a short bond and a risky asset.

There is a vast literature on ruin probability, which has been mainly studied as a problem of insurance company with classical risk models. The corner stone is laid by the Cramér–Lundberg model introduced in the early 20th century with featuring results of the Lundberg inequality and Cramér–Lundberg approximation. As discussed by Rolski et al. (1999), only some special cases such as mixed-exponential or phase-type claim size distributions allow closed form solutions for the ruin probability. Many efforts are made to study the properties of ruin probability under different setting. Recently, Albrecher and Boxma (2004) study the impacts of dependence between claim size and inter-arrival time. In their model, the previous claim size exerts its influence on the change of the inter-arrival time distribution via a threshold structure. The ruin probability is expressed as the inversion of Laplace transform of a known function form. Moreover, in a renewal risk model, Leipus and Šiaulys (2008) investigate how strongly subexponential claim sizes affect the tail behavior of the probability of ruin within finite time. Given the condition that the risk process is characterized with compound Poisson process and the expected value premium principle is utilized to measure reinsurance, Liang and Young (2018) find the optimal investment and reinsurance strategy to minimize the risk of ruin for an individual insurance company. Other recently research on ruin probability in various settings with, e.g., Markov chain interest or multivariate regularly varying claims, include: Cai and Dickson (2004), Cardoso and Waters (2005), Yang et al. (2011), Konstantinides and Li (2016), Kabanov and Pergamenschikov (2020), Liang et al. (2020), among many others. In contrast to ruin probability models of insurance companies, our model focus on the impacts of maintaining cost on charitable funds' ruin probability and wealth process. Moreover, since the time-series properties of the fixed cost and variable cost determining the ruin probability of charitable funds are fundamentally different from claims in traditional insurance models, the expressions of ruin probability in our setting yield very different intuitions.

Our work is also related to the literature of financial management of non-profit institutions and microfinance institutions. The study conducted by Calabrese and Ely (2017) contributes to a comprehensive understanding of the distinctive financial characteristics of

non-profit institutions by utilizing the new endowment data. After exploring how the firm performance and corporate governance are correlated in microfinance institutions (MFI), Mersland and Strøm (2009) conclude that non-profit organizations and shareholder firms have comparable financial performance and outreach. Gutiérrez-Nieto et al. (2009) examine the effectiveness of MFIs in terms of financial and social outputs, and they think MFIs, as specialized institutions, are supposed to play an efficient role in both financial and social aspects. Tchakoute-Tchuigoua (2010) observes that private corporations outperform non-profit NGOs if the portfolio quality is considered as one of the performance indicators. He also finds that for-profit MFIs have more social efficiency than non-profit MFIs. Regarding ownership structure, Hudon and Périlleux (2014) show that the larger part of the surplus of cooperatives is typically allocated to employees. In contrast, there is no significant difference in the way that non-profit organizations and shareholder-firm MFIs distribute earnings among their major stakeholders. For more research on non-profit institutions with different focus, e.g., impacts of financial crisis, discrimination related to microcredit officers, and credit risk, see Chen and Panjer (2009), Serrano-Cinca and Gutierrez-Nieto (2014), Labie et al. (2015), Wijesiri (2016), Guo and Peng (2020) and Hossain et al. (2020). Our research adds to this line of literature by investigating how the maintain cost affect the ruin probability of charitable fund.

Besides, our definitions of fixed cost and proportional cost differ from traditional transaction costs in trade. Rebalancing transaction costs have significant impacts on active portfolio management (Liu and Loewenstein 2002; Liu 2004; Chidambaran 2007; Chen et al. 2012; Kashif et al. 2020). In this paper, the costs are proxies of the constant cost and variable cost of maintaining a trust or fund. Few studies contribute to understanding the financial impacts of the operating fixed cost on the ruin probability of a trust or a fund. This paper provides some insights into these impacts.

The rest of the paper is organized as follows. Section 2 establishes the benchmark model and derives analytical expressions of ruin probability. Section 3 investigates the impact of maintaining costs on optimal portfolio and ruin probability. Section 4 provides ruin probability model with stochastic interest rate. Section 5 contains concluding remarks. “Appendix” offers derivation of ruin probability and related proofs.

## 2 Ruin probability with maintaining cost: benchmark model

In this section, we provide an analytically tractable model that assumes the percentage allocations to different asset classes in funds have fixed target values. The model can be used to analyze the impact of investment opportunity on ruin probability with the presence of maintaining cost.

### 2.1 The problem

Suppose a manager runs a fund and cares about the ruin probability of the fund. There are two kinds of possible costs or minimal spending to keep the fund alive: a fixed cost  $k_0$  per unit time and a variable cost  $k_1$  per unit time. The fixed cost is the constant cost including, for instance, salaries of employees, rent, management fees, administrative fees, operating costs, depreciation, property taxes and non-employment taxes, pension expense and maintenance and repair expense. The variable cost is proportional to the wealth, including the stipulated spending by rules. The manager must terminate the trust when the wealth

reaches zero. The manager cares about the probability of terminating or the probability of maintaining operability permanently.

Consider an investment opportunity of an initial endowment of  $W_0 > 0$  and possible investments in a single risky asset (a stock or an index). The process for the reinvested stock price is governed by the usual lognormal process solving,

$$\frac{dS_t}{S_t} = \mu_0 dt + \sigma dZ_t, \quad (1)$$

subject to some given initial stock price  $S_0$ , where  $\mu_0$  and  $\sigma > 0$  are constants, and  $Z$  is a standard Wiener process. Note that the price process (1) also can be interpreted as a board market index. Because specialized funds usually invest in passively managed funds tracking index. Passive managed funds are prevalent in capital markets, and many studies show that passive managed funds perform as well as or outperform active managed funds, after taking transaction costs and management fees into account (Wermers' 2000; French 2008; Fama and French 2010).

In the real world, the percentage allocations to different asset classes in funds have fixed target values or ranges. Therefore, for simplicity of analysis, in this benchmark model, we assume that the fund does not rebalance the portfolio weights, i.e., the fund invests in the risky asset and the riskless bond with fixed portfolio weights.

Assume the wealth process of the fund,  $w_t$ , reaches zero for the first time at  $\tau$ , that is  $\tau \equiv \inf(t | w_t = 0)$ . If the wealth is always positive, then  $\tau \equiv +\infty$ . The termination condition of the fund is  $w_t \leq 0$ , and we assume  $\forall t > \tau, w_t = 0$ . Hence, the termination indicator is defined as

$$I_t \equiv \begin{cases} 0, & t < \tau, \text{ i.e., } w_t > 0, \\ 1, & t \geq \tau, \text{ i.e., } w_t = 0. \end{cases} \quad (2)$$

With the definition of the termination indicator, we can state the ruin probability of the fund at a finite time to be a function of wealth as (because the value of indicator depends on if wealth is zero or not)

$$P(w) \equiv E \left[ \int_{t=0}^{\infty} dt I_t(w_t) \right], \quad (3)$$

which is not a function of time explicitly.<sup>1</sup> When wealth reaches zero, the value of the termination indicator  $I_t$  changes. The objective function indicates the incentive to minimize the probability of change in  $I_t$ . Moreover, from  $t = 0$  to  $t = +\infty$ , the integrated weight is always one. Hence, the longevity of the trust for an infinite time is as important as for any finite time, and thus this objective function models the incentive to maintain operability permanently.

Assume the variable-cost adjusted expected return is  $\mu = \mu_0 - k_1$ , and the wealth dynamic conditional on survival follows

<sup>1</sup> It is due to the assumption of infinite horizon which implies that the Bellman equation can be simplified to an ODE. Whereas, in a standard Merton model with deterministic finite horizon, the value function will be a function of time, usually leading to the Bellman equation as a PDE. Specifically, conditional on the same amount of wealth  $w$ , the value function  $P$  will be independent of time  $t$ , because let  $P(w) \equiv E[\int_0^\infty dt I_t | w_0 = w^*]$ , then for any finite  $t$ , we will also have  $E[\int_t^\infty dt I_t | w_t = w^*] = P(w)$ . To prove, just let  $v = s - t$ , then we have  $E[\int_t^\infty dt I_t | w_t = w^*] = E[\int_0^\infty dt I_v | w_0 = w^*] = P(w)$ .

$$dw_t = \mu w_t dt + \sigma w dZ_t - k_0 dt, \quad (4)$$

where  $\mu$  is the proportional-cost-adjusted expected return. In other words, by decreasing expected investment return correspondingly, the problem is reduced to a case with no proportional cost.

Note with the fixed cost  $k_0$ , the wealth does not follow a lognormal process. We cannot obtain a neat analytical solution of the SDE of the wealth process. However, as we will show, we can still achieve some neat analytical results of the probability of maintaining operability permanently, in a specific set of parameter values.

## 2.2 Boundary conditions and solution

We now solve for the ruin probability using the martingale technique pioneer by Fleming and Rishel (1975). The ruin probability at any finite time should satisfy the following ODE (see detailed derivation in “[Derivation of Bellman equation for benchmark model](#)” section in “[Appendix](#)”),

$$(\mu w_t - k_0)P_w + \frac{1}{2}\sigma^2 w^2 P_{ww} = 0. \quad (5)$$

It is easy to verify that the general solution of the ODE is given as (see “[Verifying solution and derivation of ruin probability](#)” section in “[Appendix](#)” for a proof)

$$P = C_1 \int_0^w \exp\left(-\frac{2k_0}{\sigma^2 \varpi}\right) \varpi^{-\frac{2\mu}{\sigma^2}} d\varpi + C_2, \quad (6)$$

where the constants  $C_1$  and  $C_2$  are to be determined by boundary conditions.

We have two boundary conditions. First, by the definition of maintaining operability we have the boundary condition at zero wealth as

$$P(w = 0) = 1,$$

which means that when wealth is zero, the fund is terminated, so the ruin probability is 1. Second, since the fund only invests in a risky asset whose return is not guaranteed, any finite amount of wealth can be potentially wiped out by fixed cost and bad luck of downward risk with a positive possibility. Thus, for the fund to maintain operability permanently, the wealth of the fund must converge to infinity. In other words, the probability of maintaining operability permanently is equal to the probability of infinite large wealth of the fund. Therefore, another boundary condition when wealth converges to infinity is given as

$$\lim_{w \rightarrow \infty} P(w) = 0.$$

Heuristically, we write

$$P(w = \infty) = 0.$$

This condition indicates that infinite wealth is needed to completely rule out any chance of ruin. Impose the boundary conditions to the general solution (6) and we have

$$C_1 = -\frac{1}{\int_0^\infty \exp(-2k_0/(\sigma^2 w)) w^{-\frac{2\mu}{\sigma^2}} dw} \quad \text{and} \quad C_2 = 1. \quad (7)$$

We will discuss the properties of the integral in  $C_1$  in the following section.

## 2.3 Investment opportunity analysis

### 2.3.1 Analytical solutions for ruin probability

In this subsection, we provide the condition needed for the fund to enjoy a positive probability of permanent operability. According Eq. (7), the condition needed to rule out the probability of permanent operability should make  $C_1 = 0$  and, thus, make the denominator of  $C_1$  explosive. We have the following lemma to summarize the properties of  $C_1$ .

**Lemma 1** *Given the expression of  $C_1$ , we have*

$$C_1 = \begin{cases} 0, & \text{if } \mu/\sigma^2 \leq \frac{1}{2}, \\ \text{a finite number,} & \text{if } \mu/\sigma^2 > \frac{1}{2}. \end{cases}$$

**Proof** We first prove that if  $\mu/\sigma^2 = 1/2$ , then  $C_1 = 0$ . To establish this result, we only need to prove that the denominator of  $C_1$  is explosive. Since  $\lim_{w \rightarrow \infty} \exp(-2k_0/(\sigma^2 w)) = 1$ , to prove the divergence of  $\int_0^\infty \exp(-2k_0/(\sigma^2 w)) w^{-1} dw$ , we only need to prove the divergence of  $\int_0^\infty (1/w) dw = \ln w \Big|_0^\infty = \infty$ , which is obvious. When  $\mu/\sigma^2 < 1/2$ , the integral is divergent with a higher order. Therefore, when  $\mu/\sigma^2 \leq 1/2$ ,  $C_1 = 0$ , and  $P(w) = 1$ , the fund cannot maintain operability permanently. On the other hand, according to basic properties of improper integral, the integral  $\int_0^\infty w^{-\frac{2\mu}{\sigma^2}} dw$  converges only when  $\mu/\sigma^2 > 1/2$ . Note  $\lim_{w \rightarrow \infty} \exp(-2k_0/(\sigma^2 w)) = 1$ , therefore,  $\int_0^\infty \exp(-2k_0/(\sigma^2 w)) w^{-1} dw$  converges if  $\mu/\sigma^2 > 1/2$ .  $\square$

To illustrate the result in Lemma 1 intuitively, we consider the solution of wealth process (4) which is given as

$$w_t = w_0 e^{(\mu - \frac{\sigma^2}{2})t + \sigma Z_t} - k_0 \int_{s=0}^t e^{(\mu - \frac{\sigma^2}{2})(t-s) + \sigma(Z_t - Z_s)} ds.$$

Without the fixed cost  $k_0$ , the logarithmic wealth has a drift of  $\mu - \sigma^2/2$ . If  $\mu/\sigma^2 \leq 1/2$ , that is the drift is zero or negative (with the fixed cost, the drift is negative), then the possibility of diverging to infinite wealth is zero.

Generally, we can express the probability of ruin at any finite time  $P$  as a function of cumulative distribution function of Gamma distribution. Interestingly, with  $\mu/\sigma^2 = 1$ , we have the following neat analytical solution for  $P$  as (see “[Verifying solution and derivation of ruin probability](#)” section in “[Appendix](#)” for derivation)

$$P(\tau < +\infty) = 1 - \exp\left(-\frac{2k_0}{\mu w}\right).$$

Note the probability of maintaining operability permanently  $P^s = 1 - P$ .

We summarize the above results as the following proposition.

**Proposition 1** Assume a fund invests in an index whose price process follows (1). When  $\mu/\sigma^2 > 1/2$ , the fund gains a positive probability of maintaining operability permanently  $P^s$  (probability of never failing at any finite time) and, it is given as

$$P^s(\tau = +\infty) = \frac{\int_0^w \exp\left(-\frac{2k_0}{\sigma^2 w}\right) w^{-\frac{2\mu}{\sigma^2}} dw}{\int_0^\infty \exp\left(-\frac{2k_0}{\sigma^2 w}\right) w^{-\frac{2\mu}{\sigma^2}} dw} = \frac{\int_{\frac{1}{w}}^\infty f(x; \alpha, \beta) dx}{\int_0^\infty f(x; \alpha, \beta) dx} = \int_{\frac{1}{w}}^\infty f(x; \alpha, \beta) dx, \quad (8)$$

where  $f(x; \alpha, \beta)$  is the probability density function of Gamma distribution<sup>2</sup> with shape parameter  $\alpha = 2\mu/\sigma^2 - 1$  and a rate parameter  $\beta = 2k_0/\sigma^2$ .

Particularly, when  $\mu/\sigma^2 = 1$ , we can obtain analytical expressions for the integrals in (8) and have

$$P^s(\tau = +\infty) = \exp\left(-\frac{2k_0}{\mu w}\right). \quad (9)$$

Moreover, when  $\mu/\sigma^2 \leq 1/2$ , the fund cannot maintain operability permanently.

**Proof** By (6) and (7), we immediately have the expression of the ruin probability  $P$ . By  $P^s = 1 - P$  and the probability density function of Gamma distribution, we obtain (8).  $\square$

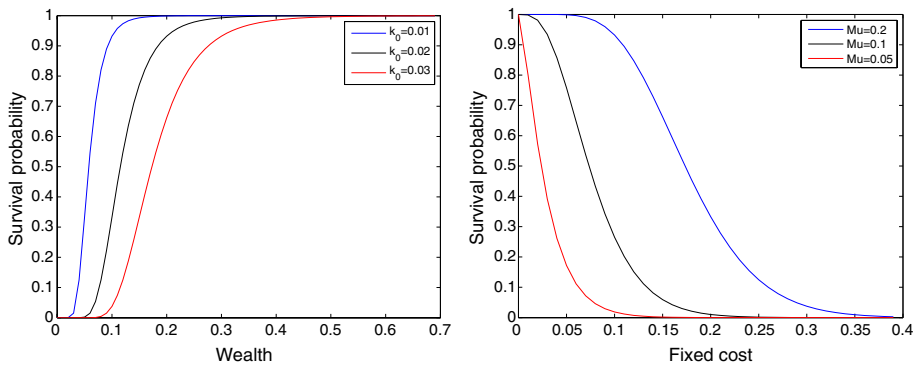
When the fixed cost is zero, the fund always survives, since wealth is positive for sure under log normal distribution. The probability of maintaining operability permanently  $P^s$  is given as one minus the cumulative distribution function of Gamma distribution, and as a result, the probability of ruin at any finite time

$$P(\tau < +\infty) = \int_0^{\frac{1}{w}} f(x; \alpha, \beta) dx = \frac{1}{\Gamma(\alpha)} \gamma(\alpha, \beta/w),$$

where  $\Gamma(\alpha)$  is the Gamma function, and where  $\gamma(\alpha, \beta/w)$  is the lower incomplete Gamma function.

The shape parameter increases with the cost-adjusted expected return, and the rate parameter increases in the fixed cost, indicating the probability of maintaining operability permanently  $P^s$  increases in cost-adjusted expected return and decreases in the fixed cost. In the special case when  $\mu/\sigma^2 = 1$ , (9) shows that the probability of maintaining operability permanently decreases exponentially with the amount of fixed cost, and increases exponentially with the amount of wealth and the cost-adjusted expected return. Moreover, both cases show that even the fund cannot invest in a riskless asset, there is still a positive probability for the fund to maintain operability permanently. Since if the wealth process has a positive drift which is large enough to offset the effect of the fixed cost, the wealth can evolve with a positive trend to infinity with a positive probability. These results and implications also hold for general cases when  $\mu/\sigma^2 \neq 1$ .

<sup>2</sup> Chapter 6 of Abramowitz and Stegun (1970) shows the detailed definition and properties of different kinds of incomplete Gamma function.



**Fig. 1** Impact of wealth and fixed cost on survival probability. The survival probability is plotted as functions of wealth and fixed cost with the following parameter values:  $\mu = 0.2$ ,  $\sigma = 0.2$  (left panel) and  $w = 1$ ,  $\sigma = 0.2$  (right panel)

### 2.3.2 Comparative statics

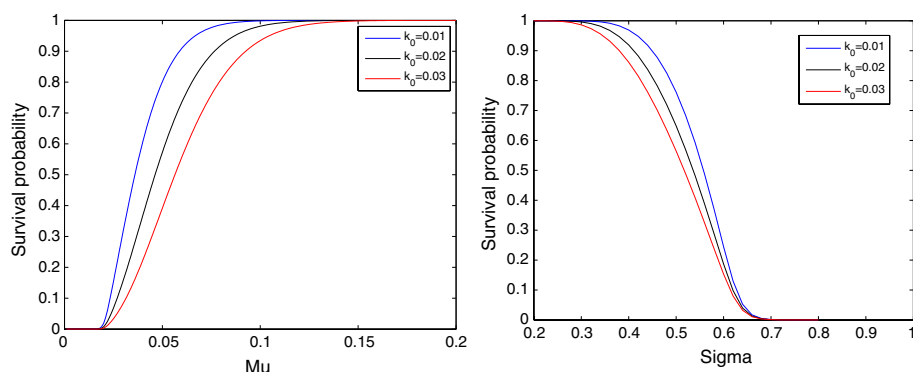
We provide comparative statics to demonstrate how investment opportunities and fixed cost affect survival probability. Figure 1 plots survival probability as function of wealth and fixed cost, respectively.

The left panel of Fig. 1 intuitively shows that survival probability increases in wealth. Interestingly, the marginal survival probability of wealth first increases and then decreases. Take the case with fixed cost being 0.03 as an example, when wealth increases from 0 to 0.08, the survival probability remains almost at zero. It is because that when wealth is small, fixed cost dominates. Any low level of wealth can be wiped out immediately. However, when wealth is higher than 0.1, the marginal survival probability of wealth increases dramatically, signaling that the impact of wealth becomes dominating. When wealth is higher than 0.3, fixed cost is well covered, hence, the survival probability approaches to one. Note that with a higher level of fixed cost, a higher level of wealth is needed to maintain the same level of survival probability. As a result, the curve shifts toward right as fixed cost increase from 0.01 to 0.03.

The right panel of Fig. 1 show that survival probability decreases in fixed cost. Particularly, when fixed cost is zero, the wealth is log normally distributed and, thus, is always positive, hence the ruin probability is zero. When the expected return is high, the marginal survival probability of fixed cost first decreases and then increases. Interestingly, when the expected return is as low as 0.05, the marginal survival probability of fixed cost always increases, echoing with the analytical results of (9) that the survival probability can be an exponential decay function of fixed cost. Moreover, a higher level of expected return is needed to cover a higher level of fixed cost to maintain the same level of survival probability. As a result, the curve shifts right as expected return increases from 0.05 to 0.2.

Figure 2 plots survival probability as function of expected return and return volatility, respectively.

The left panel of Fig. 2 shows that survival probability increases in expected return. The impact of a higher expected return is very similar to of a higher level of wealth, since expected higher return tends to give rise to more wealth. While the right panel of Fig. 2 shows that survival probability decreases in return volatility. The impact of a higher return volatility is very similar to of a higher fixed cost.



**Fig. 2** Impact of expected return and return volatility on survival probability. The survival probability is plotted as functions of expected return and return volatility with the following parameter values:  $\mu = 0.2$ ,  $w = 1$  (left panel) and  $w = 1$ ,  $\mu = 0.2$  (left panel)

### 3 Ruin probability with portfolio choices

The previous section shows that it is possible for the fund to maintain operability permanently with fixed portfolio weights. More sophisticatedly, when the fund can trade in two different stocks/indices, it might form a nontrivial portfolio choice to enhance its chance to maintain operability permanently. In this section, we establish a model in which the fund is allowed to rebalance its wealth in a portfolio of two risky assets.

#### 3.1 The model

We assume the fund can still trade in the asset  $A$  whose price process follows (1). In addition, the fund can trade in asset  $B$  which offers an extra risk premium  $\Delta$  per unit time as a compensation of extra risk caused by another Brownian motion  $Z^0$  which is independent of the Brownian motion of asset  $A$ ,  $Z$ . The dynamics of price of asset  $B$  is given as

$$\frac{dS_t^0}{S_t^0} = (\mu_1 + \Delta)dt + \sigma_1 dZ_t + \sigma_0 dZ_t^0,$$

where  $\sigma_0$ , and  $\sigma_1$  is the volatility associated to the two independent Brownian motion  $Z^0$  and  $Z_t$ , respectively, and  $\mu_1$  and  $\Delta$  are the expected returns associated with  $Z^0$  and  $Z_t$ , respectively. If let  $\mu_1 = r$  and  $\sigma = \sigma_1 = 0$ , our model is reduced to the conventional case where investors can invest in both a riskless bond and a stock (Merton 1969, 1971).

Let the fund's portfolio weight of asset  $B$  be  $\theta_t$ , taking the fixed cost into account, thus the wealth process of the fund conditional on survival follows,

$$\begin{aligned} dw &= w((1 - \theta_t)(\mu dt + \sigma dZ_t) + \theta_t((\mu_1 + \Delta)dt + \sigma_1 dZ_t + \sigma_0 dZ_t^0)) - k_0 dt \\ &= w(\mu dt + \sigma dZ_t + \theta_t \hat{\Delta} dt + \theta_t((\sigma_1 - \sigma)dZ_t + \sigma_0 dZ_t^0)) - k_0 dt, \end{aligned} \quad (10)$$

where  $\hat{\Delta} = \mu_1 - \mu + \Delta$ . The simplified dynamics of wealth indicates that the fund always invests part of the wealth in asset  $A$  and holds it fixed. Intuitively, we can possibly view

asset  $A$  as a non-traded asset, e.g., human capital. Moreover, the fund is allowed to speculate with the trade-off between the adjusted expected return  $\hat{\Delta}$  and the volatility derived from the volatility of asset  $A$  and asset  $B$ . Again, let the ruin probability of the fund at any finite time be a function of wealth as  $P(w)$ , and the definition of the termination indicator (2) can be employed to present the choice problem of maintaining operability permanently in the following form.

**Problem 1** Choose an adapted risky portfolio process  $\{\theta_t\}_{t=0}^{\infty}$  and an adapted wealth process  $\{w_t\}_{t=0}^{\infty}$  to minimize the probability of ever terminating the trust, i.e.,

$$P(w) \equiv \min_{\theta} \mathbb{E} \left[ \int_{t=0}^{\infty} dI_t \right] \\ \text{s.t. } dw_t = (1 - \iota_t) \left[ w(\mu dt + \sigma dZ_t + \theta_t(\hat{\Delta} dt + (\sigma_1 - \sigma)dZ_t + \sigma_0 dZ_t^0)) - k_0 dt \right], \quad (11) \\ w_t \geq 0, \quad \text{for } t \geq 0,$$

subject to the initial condition  $w_0 = W_0$ . It is assumed that  $\mu, \mu_1, \Delta, \sigma_0, \sigma, \sigma_1, k_0$ , and  $k_1$  are all positive.

We first provide a sketch of the deviations of the solution to Problem 1. The detailed derivations are in Sect. 4 and “Derivations of optimal portfolio and Bellman equation” section in “Appendix”. By applying the Davis–Varaiya martingale principle of optimal control (Davis and Varaiya 1973; Fleming and Rishel 1975), define  $M_t \equiv \int_{s=0}^t dI_s + P(w_t)$ , then to achieve minimum,  $M_t$  should be a martingale. Hence, the drift of  $M$  should be zero, which yields the following Bellman equation as an ODE

$$\min_{\theta_t} \left[ (w(\mu + \theta_t \hat{\Delta}) - k_0) P_w + \frac{1}{2} (\sigma^2 + 2\theta \sigma (\sigma_1 - \sigma) + \theta^2 \hat{\sigma}^2) w^2 P_{ww} \right] = 0, \quad (12)$$

where  $\hat{\sigma}^2 = \sigma_0^2 + (\sigma_1 - \sigma)^2$ . The first-order condition with respect to  $\theta_t$  gives the expression of optimal portfolio choice,

$$\hat{\Delta} w P_w + \sigma (\sigma_1 - \sigma) w^2 P_{ww} + \theta_t \hat{\sigma}^2 w^2 P_{ww} = 0 \implies \theta_t^* = -\frac{\hat{\Delta} P_w}{\hat{\sigma}^2 w P_{ww}} - \frac{\sigma (\sigma_1 - \sigma)}{\hat{\sigma}^2}. \quad (13)$$

Substitute it back to ODE (12) yields

$$0 = aw - k_0 + bw^2 \frac{P_{ww}}{P_w} - \frac{\hat{\Delta}^2 P_w}{2\hat{\sigma}^2 P_{ww}}, \quad (14)$$

where

$$a \equiv \mu - \frac{\hat{\Delta} \varphi}{\hat{\sigma}^2}, \quad b \equiv \frac{\sigma^2 \sigma_0^2}{2\hat{\sigma}^2}, \quad \phi \equiv -\frac{\varphi}{\hat{\sigma}^2}, \quad \text{and } \varphi \equiv \sigma (\sigma_1 - \sigma).$$

By solving the second order ODE (14) with the boundary conditions of

$$P(w = 0) = 1 \quad \text{and} \quad P(w = \infty) = 0,$$

we obtain the ruin probability  $P$  and thus the survival probability  $P^s = 1 - P$ . Furthermore, by (13) we obtain the optimal portfolio weight in asset  $B$ . The following Proposition 2 summarize these results.

**Proposition 2** *The optimal portfolio weight in asset B is given as a function of wealth, fixed cost, and other parameters of asset B as*

$$\theta^* = -\frac{\hat{\Delta}P_w}{\hat{\sigma}^2_w P_{ww}} - \frac{\varphi}{\hat{\sigma}^2} = \frac{1}{\hat{\Delta}} \left( \frac{k_0}{w} - a + \sqrt{\left(a - \frac{k_0}{w}\right)^2 + \frac{2b\hat{\Delta}^2}{\hat{\sigma}^2}} \right) - \frac{\varphi}{\hat{\sigma}^2}.$$

With the optimal portfolios of asset A and asset B, when  $(a + \bar{\mu})/2b > 1$ , the fund enjoys a positive survival probability and, this probability is given as

$$P^s(\tau = +\infty) = \frac{\int_0^w \left| \frac{a-x}{\varpi} \right|^{\frac{a}{2b}} \left| \frac{\bar{\mu}+x}{\bar{\mu}-x} \right|^{\frac{\bar{\mu}}{2b}} \exp\left(\frac{a^2-\bar{\mu}^2}{4b(x-a)} - \frac{k_0}{2b\varpi} - \frac{x}{4b}\right) d\varpi}{\int_0^\infty \left| \frac{a-x}{\varpi} \right|^{\frac{a}{2b}} \left| \frac{\bar{\mu}+x}{\bar{\mu}-x} \right|^{\frac{\bar{\mu}}{2b}} \exp\left(\frac{a^2-\bar{\mu}^2}{4b(x-a)} - \frac{k_0}{2b\varpi} - \frac{x}{4b}\right) d\varpi}, \quad (15)$$

where

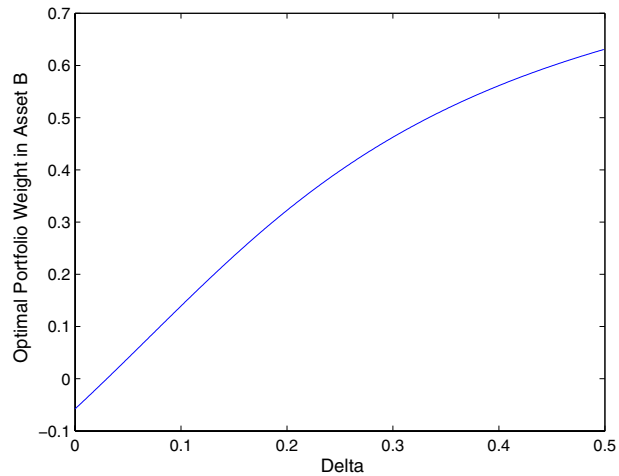
$$\begin{aligned} x &= \frac{k_0}{\varpi} - \sqrt{\frac{k_0^2}{\varpi^2} - \frac{2ak_0}{\varpi} + \bar{\mu}^2}, \quad \bar{\mu}^2 = a^2 + \frac{2b\hat{\Delta}^2}{\hat{\sigma}^2}, \\ a &= \mu - \frac{\hat{\Delta}\varphi}{\hat{\sigma}^2}, \quad b = \frac{\sigma^2\sigma_0^2}{2\hat{\sigma}^2}, \quad \phi = -\frac{\varphi}{\hat{\sigma}^2}, \quad \varphi = \sigma(\sigma_1 - \sigma), \\ \hat{\Delta} &= \mu_1 - \mu + \Delta, \quad \hat{\sigma}^2 = \sigma_0^2 + (\sigma_1 - \sigma)^2. \end{aligned}$$

Moreover, when  $(a + \bar{\mu})/2b \leq 1$ , the fund cannot maintain operability forever.

**Proof** See Sect. 4 for the verification theorem and see “Proof of Proposition 2” and “Derivations of optimal portfolio and Bellman equation” sections of “Appendix” for related derivations.

The portfolio is endogenized as a function of wealth, and it is not an explicit function of time. With similar argument in the previous section, and note that by Lemma 2, we have that as wealth goes to infinity,  $(\bar{\mu} + x)/(\bar{\mu} - x) \sim 1/\varpi$  (see Lemma 2 and its proof in “Lemma 2 and Proof” section in “Appendix”), we obtain that when  $(a + \bar{\mu})/2b \leq 1$ , the integral in denominator is divergent. Hence, the fund cannot maintain operability forever. Note that  $\bar{\mu} > a$ , a large enough absolute value of  $\hat{\Delta}$  and thus a large enough  $\bar{\mu}$  will give rise to a positive chance of survival permanently. This can come from either a large positive expected return or a small expected return of asset B. Because the fund can increase the survival probability by taking negative position in asset B. For instance, if  $\mu_1$  is very small,  $\hat{\Delta}^2$  could be large, and thus  $\bar{\mu}$  is large. Hence as long as the absolute value of  $\hat{\Delta}$  is large enough, the fund has a good chance to survive permanently. This result is also demonstrated by the comparative statics analysis in the following Sect. 3.3. Particularly, when  $\mu_1 = \mu$ , and  $\sigma_1 = \sigma$ , we have  $\bar{\mu} > \mu$ . Therefore, comparing to the results in Proposition 1, the condition needed for maintaining operability permanently is easier to be satisfied and less restrictive, indicating that the additional asset does enhance the survival probability.  $\square$

**Fig. 3** Impact of additional expected return on optimal portfolio weight. The optimal portfolio weight is plotted as a function of additional expected return with the following parameter values:  $\mu_1 = 0.1$ ,  $\sigma_1 = 0.15$ ,  $\mu = 0.25$ ,  $k_0 = 0.04$ ,  $\sigma = 0.2$ ,  $\sigma_0 = 0.2$ , and  $w = 1$



### 3.2 Properties of optimal portfolio

First of all, we plot the optimal portfolio weight in asset  $B$  as a function of the additional expected return  $\Delta$  in asset  $B$ .

Figure 3 shows that the portfolio weight in asset  $B$  increases in additional expected return, since the larger value of additional expected return leads to higher total expected return and Sharpe ratio. In particular, when additional expected return is small, the optimal investment in asset  $B$  can be negative. This is because the adjusted expected return  $\hat{\Delta}$  is negative, a short position in asset  $B$  is profitable. Moreover, we can also see that the optimal portfolio weight in asset  $B$  is a concave function of the additional expected return. It is because that as the additional expected return increases to a high level, the ruin probability becomes very low and close to zero. In this case, there is not much room for improvement in survival probability by increasing the portfolio weight in asset  $B$ , given the volatility unchanged. Therefore, the marginal portfolio weight in asset  $B$  decreases in  $\Delta$ .

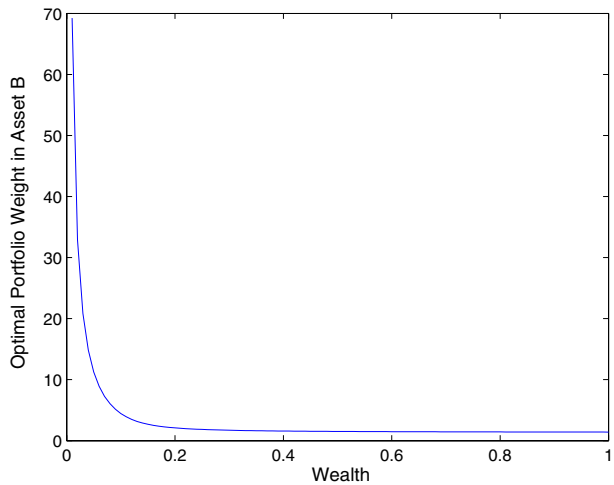
We plot the optimal portfolio weight in asset  $B$  as a function of wealth,  $w$ .

Figure 4 shows the optimal portfolio weight in asset  $B$  decreases in the level of wealth. The intuition is that when the wealth level is low, the fund faces large ruin probability due the fixed maintaining cost. Hence, the manager has a stronger incentive to gamble and earn extra risk premium from asset  $B$ , which yields additional expected return by bearing additional risk. Notably, the existence of fixed cost drives the portfolio weight in asset  $B$  to infinity as the wealth level converges to zero.

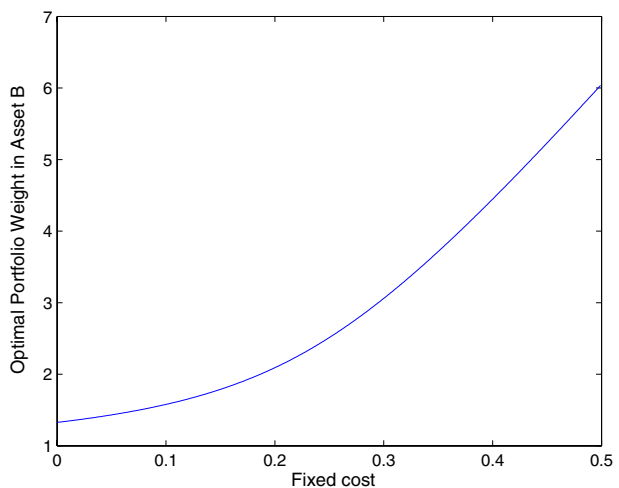
We plot the optimal portfolio weight in asset  $B$  as a function of the fixed cost,  $k_0$ .

Figure 5 shows the optimal portfolio weight in asset  $B$  increases in the fixed maintaining cost. Intuitively, a higher level of fixed maintaining cost deprives more wealth per unit time, leading to a higher ruin probability. To overcome the increasing fixed spending, the fund tends to take more risk and earn additional premium to increase the survival probability. Hence, it is the incentive to gamble and earn more premium that gives rise to a higher portfolio weight in asset  $B$ . Moreover, the optimal portfolio weight is a convex function of the fixed cost. Because, when the fixed cost is low, the wealth and the expected return from asset  $A$  can well cover it. Therefore, a bit increase in fixed cost would not have great impact on ruin probability, leading to low marginal

**Fig. 4** Impact of wealth on optimal portfolio weight. The optimal portfolio weight is plotted as a function of wealth with the following parameter values:  $\mu_1 = 0.14$ ,  $\sigma_1 = 0.15$ ,  $\mu = 0.15$ ,  $k_0 = 0.04$ ,  $\sigma = 0.2$ ,  $\Delta = 0.12$ , and  $\sigma_0 = 0.1$



**Fig. 5** Impact of fixed cost on optimal portfolio weight. The optimal portfolio weight is plotted as a function of fixed cost with the following parameter values:  $\mu_1 = 0.14$ ,  $\sigma_1 = 0.15$ ,  $\mu = 0.15$ ,  $\sigma = 0.2$ ,  $\Delta = 0.12$ ,  $\sigma_0 = 0.1$ , and  $w = 1$

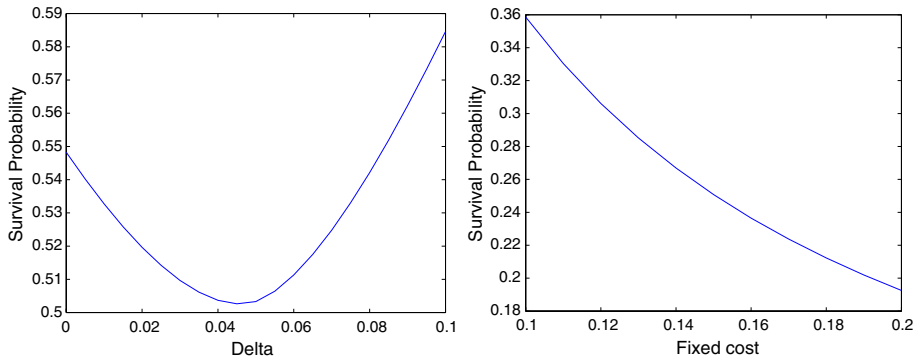
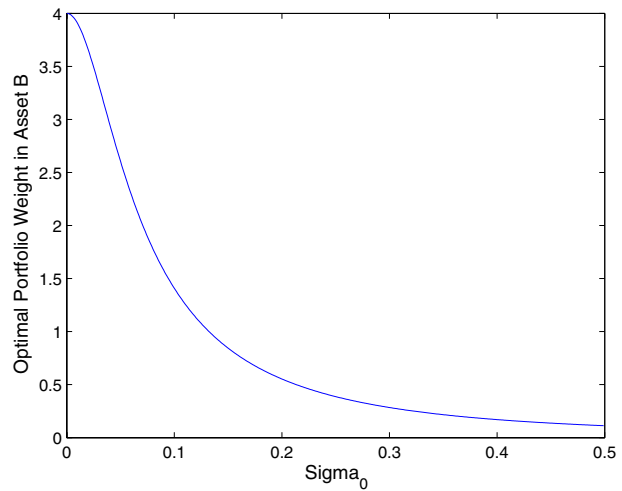


change in the optimal portfolio weight in asset  $B$ . However, when the fixed cost is high, marginal increase in fixed cost significantly increase the ruin probability, leading to a larger marginal change in the optimal portfolio in asset  $B$ .

We plot the optimal portfolio weight in asset  $B$  as a function of return volatility of asset  $B$ ,  $\sigma_0$ .

Figure 6 shows that the optimal portfolio weight in asset  $B$  decreases in one of the return volatility of asset  $B$ , i.e.,  $\sigma_0$ . This return volatility is associated with the shock unshared by asset  $A$ . Intuitively, when the expected return is constant, increase in volatility will decrease asset  $B$ 's Sharpe ratio and make asset  $B$  less attractive, since asset  $A$  remains unchanged. Therefore, the investor tends to invest more wealth in asset  $A$ .

**Fig. 6** Impact of return volatility associated with asset  $B$ 's unique shock on optimal portfolio weight. The optimal portfolio weight is plotted as a function of the return volatility  $\sigma_0$  of Asset  $B$  with the following parameter values:  $\mu_1 = 0.14$ ,  $\sigma_1 = 0.15$ ,  $\mu = 0.15$ ,  $k_0 = 0.04$ ,  $\Delta = 0.12$ ,  $\sigma = 0.2$ , and  $w = 1$

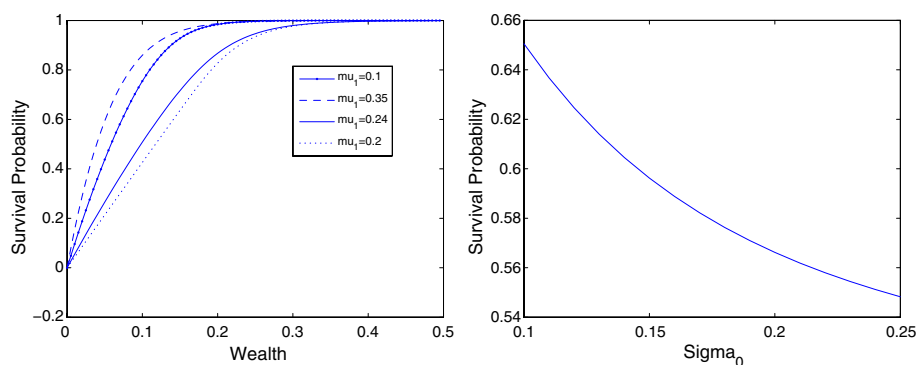


**Fig. 7** Impact of additional expected return and fixed cost on survival probability. The survival probability is plotted as a function of additional expected return with the following parameter values:  $\mu_1 = 0.2$ ,  $\sigma_1 = 0.15$ ,  $\mu = 0.25$ ,  $k_0 = 0.04$ ,  $\sigma = 0.2$ ,  $\sigma_0 = 0.25$ , and  $w = 0.12$ . The survival probability is plotted as a function of fixed cost with the following parameter values:  $\mu_1 = 0.14$ ,  $\sigma_1 = 0.15$ ,  $\mu = 0.25$ ,  $\sigma = 0.2$ ,  $\Delta = 0.06$ ,  $\sigma_0 = 0.25$ , and  $w = 0.12$

### 3.3 Properties of survival probability

We plot the survival probability as functions of the additional expected return,  $\Delta$ , and fixed cost,  $k_0$ , respectively.

The left panel in Fig. 7 shows that the survival probability, i.e., one minus the ruin probability, may not always increases in the additional expected return. This is because when the investor takes short position in the asset  $B$ , a lower  $\Delta$  will make the short position more profitable, leading to a higher survival probability. However, when  $\Delta$  is larger than 0.05, the investor will take long position in asset  $B$ . In this case, higher  $\Delta$  implies that investment in asset  $B$  is more profitable and is better at keeping the wealth away from zero. The right panel in Fig. 7 plots the survival probability as a function of the fixed cost,  $k_0$ . Not surprisingly, the survival probability decreases in fixed cost, which



**Fig. 8** Impact of wealth and additional volatility on survival probability. The survival probability is plotted as a function of fixed cost with the following parameter values:  $\mu = 0.25$ ,  $\sigma_1 = 0.25$ ,  $\sigma = 0.2$ ,  $\Delta = 0.06$ ,  $\sigma_0 = 0.15$ ,  $k_0 = 0.04$ . The survival probability is plotted as a function of the additional volatility with the following parameter values:  $\mu_1 = 0.14$ ,  $\sigma_1 = 0.15$ ,  $\mu = 0.25$ ,  $k_0 = 0.04$ ,  $\Delta = 0.06$ ,  $\sigma = 0.2$ , and  $w = 0.12$

drives the wealth down. Moreover, we plot the survival probability as functions of the level of wealth,  $w$ , and the additional return volatility,  $\sigma_0$ , in the following figure.

The left panel in Fig. 8 shows that the survival probability increases with the level of wealth. Due to our assumption on positive  $P_{ww}$ , the marginal survival probability at zero wealth is positive, this forms a contrast to the case in Sect. 2 where the marginal survival probability at zero wealth is zero. Moreover, we change the value of  $\mu_1$  to see the impact of different level of the adjusted expected return  $\hat{\Delta}$ . We take the case with  $\mu_1 = 0.24$  as a benchmark. We can see from the left panel that allowing short position in asset  $B$  will increase survival probability when  $\mu_1$  is low enough. With  $\mu_1$  as low as 0.1,  $\hat{\Delta}$  is negative, and the investor shorts asset  $B$ . This investment strategy is profitable and increases the survival probability relative to the case with  $\mu_1 = 0.24$ . Of course, when  $\mu_1 = 0.35$ , which means the expected return is much higher than the benchmark. The investor will take a long position in asset  $B$  and the survival probability is much improved, leading to the highest survival probability of all four cases. When  $\mu_1 = 0.2$ , the expected return is lower, but not small enough to take a short position. Therefore, the investor suffers a long position in asset  $B$  with lower expected return, leading to the lowest survival probability in the four cases. This form a contrast to the case of  $\mu_1 = 0.1$ , which is even lower and incurs a short position. This result is consistent with the implication of Proposition 2. Finally, we plot the survival probability as a function of the additional return volatility  $\sigma_0$  in the right panel of Fig. 8, we find that the survival probability decreases in  $\sigma_0$ , which is a very intuitive result.

## 4 Verification theorem for Proposition 2

We now turn to a formal proof of the optimality of the portfolio strategy in Proposition 2, which can be considered to be a verification theorem, because it verifies that the solution to the Bellman equation as presented in the informal derivation earlier is actually optimal. We first prove three lemmas, i.e., Lemmas 2, 3 and 3. Employing the three lemmas, we prove the optimality at the end of this section. Define

$$M_t \equiv \int_{s=0}^t dt_s + P(w_t, t_t), \quad (16)$$

for  $t > 0$ , where  $P(\cdot, \cdot)$  is the proposed value function defined as  $P = 1 - P^s$ , and  $P^s$  is given as (15) in the statement of Proposition 2.<sup>3</sup> This denotes the probability of subsequent termination in the future after date  $t$ . For the optimal strategy,  $M_t$  should be the expected value of the problem, conditional on the information available at time  $t$ , as long as  $M_t$  is a martingale for  $t \in [0, \infty)$  and the residual part converges to 0 ( $\lim_{t \rightarrow \infty} E[P(w_t, t_t)] = 0$ ). Any other strategy is known to do worse if it makes  $M_t$  a submartingale with a residual part in the expectation larger than or equal to zero ( $\lim_{t \rightarrow \infty} E[P(w_t, t_t)] \geq 0$ ). The Bellman equation is the condition for  $M_t$  to become a driftless diffusion process, i.e., a local martingale, for the optimal strategy and a local submartingale for all strategies. The proofs employ some properties of bounded processes to show that the bounded local submartingales and the bounded martingale are submartingales and martingales, respectively. The following proof shows that  $M_t$  is a local submartingale for all strategies and is a local martingale for the proposed optimum.

**Lemma 2** *For all feasible strategies, the process  $M_t$  defined in (16) is a local submartingale. For the proposed optimal strategy,  $M_t$  is a local martingale.*

**Proof** Note that by the dynamics of wealth (10) and by Itô's Lemma (letting  $dt^2 = dt dZ = 0$ ,  $dZ^2 = dt$ ), we have

$$\begin{aligned} dM_t &= dt_t + P_w dw_t + \frac{1}{2} P_{ww} (dw_t)^2 + (P(w_t, 1) - P(w_t, 0)) dt_t \\ &= dt_t + (1 - t_t) P_w (w(\mu dt + \sigma dZ_t + \theta(\hat{\Delta} dt + (\sigma_1 - \sigma) dZ_t + \sigma_0 dZ_t^0)) - k_0 dt) \\ &\quad + \frac{1}{2} (1 - t_t)^2 P_{ww} w^2 (\sigma^2 + 2\theta\sigma(\sigma_1 - \sigma) + \theta^2 \hat{\sigma}^2) dt + (P(w_t, 1) - P(w_t, 0)) dt_t, \end{aligned}$$

where we employ

$$\begin{aligned} (dw)^2 &= (w(\mu dt + \sigma dZ_t + \theta(\hat{\Delta} dt + (\sigma_1 - \sigma) dZ_t + \sigma_0 dZ_t^0)) - k_0 dt)^2 \\ &= w^2 (\theta^2 \hat{\sigma}^2 + 2\theta(\sigma_1 - \sigma)\sigma + \sigma^2) dt. \end{aligned}$$

Note that  $M_t$  should be a martingale, hence, the drift of  $M$  should be zero, which yields the following Bellman equation as an ODE

$$\min_{\theta_t} \left[ 0 = (w(\mu + \theta_t \hat{\Delta}) - k_0) P_w + \frac{1}{2} (\sigma^2 + 2\theta\sigma(\sigma_1 - \sigma) + \theta_t^2 \hat{\sigma}^2) w^2 P_{ww} \right] = 0. \quad (17)$$

First-order condition with respect to  $\theta_t$  yields the optimal portfolio choice as (13). Substituting of the optimal choice of  $\theta_t$  into (17) yields the Bellman equation which implies a

<sup>3</sup> The value of indicator  $t_t$  depends on if wealth  $w$  is zero or not. Hence, essentially  $P$  and  $P^s$  are just functions of wealth. The presentation of two-variable function  $P(w_t, t_t)$  is used mainly in the proof of verification theorem when applying Itô's Lemma to highlight that the ruin probability  $P$  depends on the indicator which jumps from zero to one when wealth reaches zero. Since in most parts of the paper, we focus on the nontrivial case of  $t_t = 0$ , i.e., the fund is alive with positive wealth. Therefore, we simply omit the condition  $t_t = 0$ , and write  $P(w, t_t = 0)$  as  $P(w)$ . Moreover, note that when indicator  $t_t = 1$ , the ruin probability  $P$  is trivial and is always one.

zero drift of  $M_t$  (the detailed derivations of the expressions of optimal portfolio choice and the Bellman equation are attached in “[Derivations of optimal portfolio and Bellman equation](#)” section in the “[Appendix](#)”). Furthermore, the proposed  $\theta_t$  can be easily verified as the minimal one, since the second derivative of the portfolio is positive  $\hat{\sigma}^2 w^2 P_{ww}$  and thus the drift is convex with respect to  $\theta_t$ . Note that we have to assume  $P_{ww} > 0$  to ensure that it is a minimization problem. Hence, all other possible strategies, the drift is either zero or positive. Because a driftless diffusion process is a local martingale, as a result, we have shown that  $M_t$  is a local submartingale for every feasible strategy and a local martingale for the proposed optimal strategy.  $\square$

We can achieve Lemma 3 based on Lemma 2.

**Lemma 3** *For feasible trading strategies,  $M_t$  is a submartingale. Moreover, given the proposed optimal strategy,  $M_t$  is a martingale.*

**Proof** Note that for all kinds of strategies,

$$0 \leq M_t = \int_{s=0}^t dl_s + P(w_t, t_t) \leq 2,$$

thus  $M_t$  is a submartingale for all feasible strategies because a local submartingale that is bounded from above is a submartingale. Note that every bounded local martingale is a martingale, thus this proves that  $M_t$  is a martingale, given the proposed optimal strategy.  $\square$

**Lemma 4** (Transversality condition) *For all feasible strategies,  $\lim_{t \rightarrow \infty} E[P(w_t, t_t)] \geq 0$ . For the proposed optimal strategy,  $\lim_{t \rightarrow \infty} E[P(w_t, t_t)] = 0$ .*

**Proof** Every trading strategy is an optimal strategy, if terminated at a finite time, i.e.,  $\exists t_T$ , such that  $t_{t_T} = 1$ . After termination, there will be no subsequent termination in the future and, thus, the probability of subsequent termination is 0. Hence, we have  $\lim_{t \rightarrow \infty} E[P(w_t, t_t)] = 0$ . We now prove that for an optimal strategy, if there is never termination,  $\lim_{t \rightarrow \infty} E[P(w_t, t_t)] = 0$ .

Note, as  $t \rightarrow \infty$ , if there is never termination, we need to have  $w_t \rightarrow \infty$  with probability 1,  $\forall X > 0$ ,  $\Pr(w_t > X) \rightarrow 1$  as  $t \rightarrow \infty$ . Since given no investment opportunity in riskless asset, with only risky investments, any finite wealth over an infinite horizon cannot guarantee never termination. With only risky assets, the fund will sooner or later encounter a large negative shock (with probability one), which could nearly wipe out the fund, and the fixed cost will exhaust the fund. Specifically, with never termination, wealth will approach to infinity with an asymptotic stable growth rate of log wealth as

$$\mu + \hat{\theta} \hat{\Delta} - \frac{1}{2} \left( \hat{\theta}^2 \sigma_0^2 + \hat{\theta}^2 (\sigma_1 - \sigma)^2 + \sigma^2 \right) > 0 \quad (18)$$

where

$$\hat{\theta} \equiv \lim_{t \rightarrow \infty} \theta^* = \frac{1}{\hat{\Delta}} \left( \sqrt{a^2 + \frac{2b\hat{\Delta}^2}{\hat{\sigma}^2}} - a \right) - \frac{\varphi}{\hat{\sigma}^2},$$

and the condition (18) is needed for our model to be well posed. One can see that by solving the following SDE with the optimal choice of portfolio:

$$\begin{aligned} dw_t &= w(\mu dt + \sigma dZ_t + \theta^*(\hat{\Delta} dt + (\sigma_1 - \sigma)dZ_t + \sigma_0 dZ_t^0)) - k_0 dt \\ &= w\left(\mu dt + \theta^* \hat{\Delta} dt - \frac{k_0}{w} dt + \sigma dZ_t + \theta^*(\sigma_1 - \sigma)dZ_t + \theta^* \sigma_0 dZ_t^0\right) \end{aligned}$$

and obtain

$$w = w_0 \exp \left( \int_{v=0}^t \left( \mu + \theta^* \hat{\Delta} - \frac{k_0}{w} - \frac{1}{2} \left( \sigma^2 + (\theta^*(\sigma_1 - \sigma))^2 + (\theta^*)^2 \sigma_0^2 \right) \right) dt + \int_{v=0}^t \sigma dZ_t + \int_{v=0}^t \theta^* (\sigma_1 - \sigma) dZ_t + \int_{v=0}^t \theta^* \sigma_0 dZ_t^0 \right)$$

Therefore, asymptotically, the wealth is expected to grow in time exponentially as a geometric Brownian motion<sup>4</sup> with

$$E[w] = w_0 \exp \left( \left( \mu + \theta^* \Delta - \frac{1}{2} \left( \sigma^2 + (\theta^*(\sigma_1 - \sigma))^2 + (\theta^*)^2 \sigma_0^2 \right) \right) t \right).$$

Given  $w_t \rightarrow \infty$  with probability 1,  $P^s \rightarrow 1$  with probability 1 by (15), since the numerator coverages to the denominator with probability 1. Therefore, by  $P = 1 - P^s$ ,  $P \rightarrow 0$  with probability 1 as time evolves to infinity and  $\lim_{t \rightarrow \infty} E[P(w_t, t_t)] = 0$ .  $\square$

**Proof of Proposition 2** We want to show that the proposed solution is optimal. If terminated at a finite time  $t_T$ , i.e.,  $t_{T^*} = 1$ , then all strategies result in an identical objective value. Therefore, all the strategies are optimal. If there is never termination, by Lemmas 3 and 4  $M_t$  is a martingale and  $\lim_{t \rightarrow \infty} E[P(w_t, t_t)] = 0$ , for the proposed optimal strategy. Therefore,

$$\begin{aligned} P(w_0, t_0) &= M_0 = \lim_{t \rightarrow \infty} E[M_t] = \lim_{t \rightarrow \infty} E \left[ \int_{s=0}^t dI_s + P(w_t, t_t) \right] \\ &= \lim_{t \rightarrow \infty} E \left[ \int_{s=0}^t dI_s \right] + \lim_{t \rightarrow \infty} E[P(w_t, t_t)] = \lim_{t \rightarrow \infty} E \left[ \int_{s=0}^t dI_s \right], \end{aligned}$$

<sup>4</sup> It is a well-known result that geometric Brownian motion tends to increase to infinity with probability one with the right set of parameters. Since if

$$w_t = w_0 \exp \left[ \int_{s=0}^t \left( \mu - \frac{1}{2} \sigma^2 \right) ds + \sigma Z_t \right].$$

then  $\log(w_t) \sim \Phi(\log(w_0) + (\mu - \sigma^2/2)t, \sigma^2 t)$ . For any fixed  $X > 0$ ,

$$\Pr(w_t \leq X) = \Pr(\log(w_t) \leq \log(X)) = \Phi \left( \frac{\log(X) - \log(w_0) - (\mu - \sigma^2/2)t}{\sigma \sqrt{t}} \right),$$

where  $\Phi(\cdot)$  is the standard normal cumulative distribution function. Therefore,

$$\lim_{t \rightarrow \infty} \Pr(w_t \leq X) = \begin{cases} 0 & \text{if } \mu - \sigma^2/2 > 0 \\ 1/2 & \text{if } \mu - \sigma^2/2 = 0 \\ 1 & \text{if } \mu - \sigma^2/2 < 0 \end{cases}.$$

which confirms that  $P(w_0, t_0)$  is the value function for the proposed optimum.

For any feasible strategy,  $M_t$  is a submartingale by Lemma 3. Furthermore, we know that  $\lim_{t \rightarrow \infty} E[P(w_t, t_t)] \geq 0$  by Lemma 4. Therefore,

$$\begin{aligned} P(w_0, t_0) = M_0 &= \lim_{t \rightarrow \infty} E[M_t] = \lim_{t \rightarrow \infty} E \left[ \int_{s=0}^t dI_s + P(w_t, t_t) \right] \\ &= \lim_{t \rightarrow \infty} E \left[ \int_{s=0}^t dI_s \right] + \lim_{t \rightarrow \infty} E[P(w_t, t_t)] \geq \lim_{t \rightarrow \infty} E \left[ \int_{s=0}^t dI_s \right], \end{aligned}$$

which proves that no strategy can do better than the proposed optimum, which is therefore optimal.  $\square$

## 5 Ruin probability with stochastic interest rate

In the previous sections, we assume the fund can invest in either one risky asset or two risky assets. These assumptions echo with the stochastic investment environment in the bond market, where the nominal interest rate varies substantially due to the fluctuation in the inflation level. One may wonder if the fund can maintain operability forever with a stochastic interest rate. This motivates us to investigate the ruin probability in a model with stochastic nominal riskless interest rate.

When consider an investment model with stochastic riskless interest, it is easy to encounter an optimal problem with two state variables, i.e., the wealth and the riskless rate [Korn and Kraft (2002), Munk and Sørensen (2004) and Chapter 2 in Rogers (2013)]. This kind of problem usually leads to Bellman equations which are PDE and hard to cope with. In the following subsection, we avoid this difficulty by modelling price processes in real terms.

### 5.1 The model

Based on the historical data of real interest rate, it is reasonable to assume the real interest rate as a constant. We assume the nominal interest rate is determined by the Fisher equation as

$$r_t = r^R + \bar{t}_t,$$

where  $r^R$  is the constant real rate and  $\bar{t}$  is the inflation. We assume the price level follows

$$\frac{dI_t}{I_t} = \bar{t}_t dt + \sigma dZ_t = (r_t - r^R) dt + \sigma dZ_t,$$

where the  $\sigma$  is the constant volatility of the price level. Moreover, the nominal value of the short bond follows

$$\frac{dB_t}{B_t} = r_t dt,$$

which gives rise to

$$B_t = B_0 \exp \left( \int_{s=0}^t r_s dt \right).$$

Note that the real value of any investment is given by nominal value divided by the price level. Therefore, by Itô's Lemma the dynamics of real value of the short riskless bond is given as

$$\begin{aligned} d(B_t/I_t) &= \frac{\partial(B_t/I_t)}{\partial B_t} dB_t + \frac{\partial(B_t/I_t)}{\partial I_t} dI_t + \frac{1}{2} \frac{\partial^2(B_t/I_t)}{(\partial I_t)^2} (dI_t)^2 dt \\ &= \frac{B_t r_t}{I_t} dt - \frac{B_t}{I_t^2} I_t ((r_t - r^R) dt + \sigma dZ_t) + \frac{1}{2} \frac{2B_t}{I_t^3} I_t^2 \sigma^2 dt, \end{aligned}$$

thus, we have

$$\frac{d(B_t/I_t)}{B_t/I_t} = (r^R + \sigma^2) dt - \sigma dZ_t. \quad (19)$$

Therefore, the real value of the short bond follows a geometric Brownian motion.

## 5.2 Investment opportunity analysis

Note that the real value of short bond actually follows a form of stock as in (1). Hence, the problem when fund invests only in a short bond or addition to a stock is essentially the same as the problems in the previous sections. Consequently, by Proposition 1, we can see even with stochastic interest rate, the fund still gains a positive survival probability by investing only in the short bond if the following condition establishes,

$$\frac{r^R + \sigma^2}{\sigma^2} = 1 + \frac{r^R}{\sigma^2} > \frac{1}{2}.$$

Moreover, the survival probability can be calculated according to Proposition 1.

Note that drift of the real value of the short bond increases with the volatility of the price level. The price level volatility will play an important role in determining the survival probability. By Proposition 2, we can see if the fund can invest in the short bond and a stock whose real value follows

$$\frac{dS_t^0}{S_t^0} = (r^R + \sigma^2 + \Delta) dt - \sigma dZ_t + \sigma_0 dZ_t^0, \quad (20)$$

then the fund always has a positive survival probability if

$$\frac{(r^R + \sigma^2) + \sqrt{(r^R + \sigma^2)^2 + \sigma^2 \Delta^2 / \sigma_0^2}}{\sigma^2} > 1.$$

Note that here we assume  $\mu_1 = \mu$  and  $\sigma_1 = \sigma$ . Interestingly, this condition always establishes due to

**Table 1** Assumptions on distribution of input

| Input                                  | Distribution | Minimum | Most likely | Maximum |
|----------------------------------------|--------------|---------|-------------|---------|
| Real rate $r^R$                        | BetaPERT     | − 0.005 | 0.018       | 0.04    |
| Price level volatility $\sigma$        | BetaPERT     | 0.001   | 0.050       | 0.1     |
| Additional expected return $\Delta$    | BetaPERT     | 0.05    | 0.175       | 0.3     |
| Additional stock volatility $\sigma_0$ | BetaPERT     | 0.05    | 0.225       | 0.4     |
| Fixed cost $k_0$                       | BetaPERT     | 0.01    | 0.045       | 0.08    |

$$\sqrt{(r^R + \sigma^2)^2 + \sigma^2 \Delta^2 / \sigma_0^2} > r^R.$$

The survival probability can be calculated by applying Proposition 2. Note that the stock price process (20) is quite specific: the expected return increases with constant volatility of the price level,  $\sigma$ , and the stock price is also driven by the shock of the price level. The intuition is that investors require compensation for bearing risk from price level, hence they ask for higher risk premium as volatile of price level becomes higher. We summarize the above results in the following Corollary.

**Corollary 1** *Assume the fund can invest in a short bond whose real value follows (19).*

- (a) *If the fund can only invest in the short bond, the fund can maintain operability forever with a positive probability when  $r^R > -\sigma^2/2$ .*
- (b) *The fund always enjoys a positive survival probability if it is allowed to invest in both the short bond and a particular risky asset whose price process follows (20).*

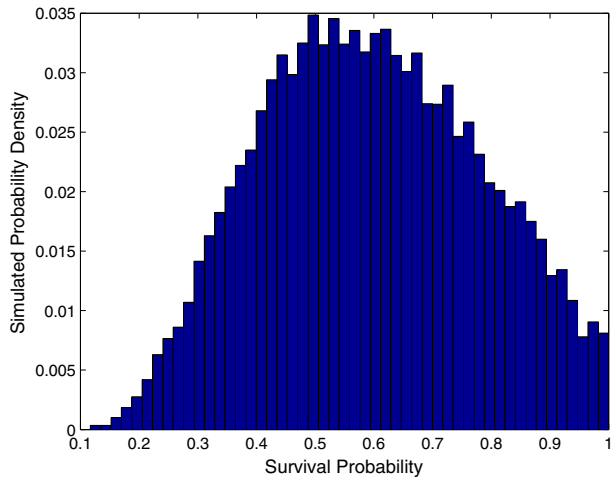
Note that we model a negative correlation between the shock in the price level and the real stock return. This assumption is supported by many researches with empirical evidence on the significantly negative correlation between inflation and stock returns with post-war data for the US and other industrialized countries, e.g. Fama and Schwert (1977), Gultekin (1983). Moreover, Day (1984) provides a theoretical framework to endogenize the negative correlation.

### 5.3 Sensitivity analysis

The permanent survival probability is affected by the uncertainty of the input parameters. To assess the impact of parameter uncertainty and obtain a confidence interval for the permanent survival probability, we perform a sensitivity analysis using the Monte Carlo simulations in the scenario with stochastic interest rate. Specifically, we treat the permanent survival probability as a function of the random parameters which follow BetaPERT distribution.<sup>5</sup> The BetaPERT distribution is symmetric and is comparable to a normal

<sup>5</sup> The betaPERT distribution is often employed in sensitivity analysis. It is a continuous distribution. It describes a situation where the minimum, maximum, and most likely values to occur are known. It is similar to the triangular distribution, except the curve is smoothed to reduce the importance of peak. The parameters of the distribution are Minimum, Most Likely, Maximum.

**Fig. 9** Distribution of permanent survival probability with 20,000 simulations



**Table 2** The simulation statistics

| Name     | Statistics | Percentiles (%) | Statistics |
|----------|------------|-----------------|------------|
| Minimum  | 0.11       | 1.00            | 0.22       |
| Maximum  | 1          | 5.00            | 0.30       |
| Mean     | 0.60       | 15.00           | 0.39       |
| Variance | 0.03       | 30.00           | 0.48       |
| St. dev. | 0.19       | 50.00           | 0.59       |
| Skewness | 2.26       | 70.00           | 0.70       |
| Kurtosis | 0.08       | 85.00           | 0.81       |
|          |            | 95.00           | 0.91       |
|          |            | 99.00           | 0.97       |

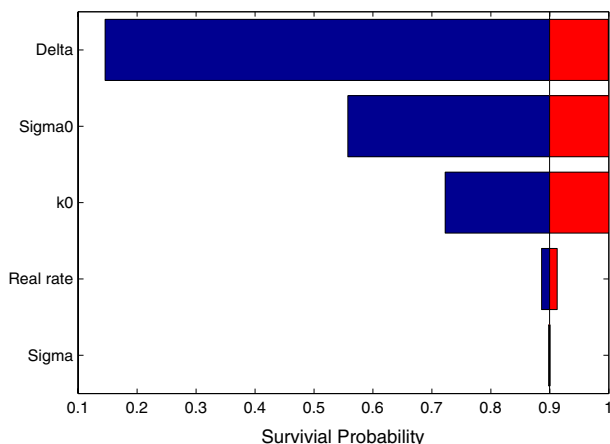
distribution with constraints on the minimum and maximum values. The following Table 1 provide the parameters of distributions for each input.

Monte Carlo simulations are very useful in determining impact of randomness of parameters. We perform simulations with 20,000 random sample for each of the 5 inputs. The implied permanent survival probability distribution is given in Fig. 9. The simulation statistics of survival probability are provided in the Table 2.

The expected permanent survival probability is 0.60, with a 95% confidence interval of 0.22 to 0.91. Moreover, the simulation results are consistent with Corollary 1 that the fund always enjoys a positive survival probability if it is allowed to invest in both the short bond and a particular risky asset (20). Moreover, we also provide the tornado diagram as Fig. 10 to demonstrate the parameter sensitivity of survival probability. Specifically, we hold all other inputs at their expected value, except one at its three different values: the minimum, the mean (base), and the maximum. Then we calculate the survival probability, and the process is repeated for each input. This allows testing the sensitivity/risk associated with one input.

As we can see from Fig. 10, survival probability is the most sensitive to the change in the additional expected return  $\Delta$  and the additional volatility of stock return  $\sigma_0$ . Therefore, when estimating input values, one should pay more attention to them to obtain more

**Fig. 10** Tornado diagram for permanent survival probability (sensitivity analysis)



accurate estimations of survival probability. Moreover, the survival probability is also considerably sensitive to the change in the fixed cost  $k_0$ .

## 6 Conclusion

We study the ruin probability of a charitable trust or fund and examine how the proportional cost and fixed cost affect the probability of ruin at any finite time when the objective is to keep a fund operable permanently. We find that with conventional investment strategies featured by fixed targeted portfolio weights, as long as the ratio of cost-adjusted expected return to return volatility is larger than half, the fund enjoys a positive possibility of maintaining operability permanently. If allowing portfolio rebalance between two risky assets (one of the asset offers extra expected return as a compensation for bearing additional risk), then the optimal portfolio selection potentially increases the survival probability. Moreover, intuitively, the optimal portfolio weight in the asset with additional expected return decreases with the level of additional return volatility and wealth, and increases with the additional expected return, the fixed maintaining cost and the return volatility of asset with no additional expected return.

We applied the results and propositions obtained in the two previous models to a case with stochastic nominal interest rate due to randomness in inflation rate, and find that under favorable investment opportunities, the fund enjoys a positive probability of maintaining operability permanently with only investment in the short bond. Moreover, with an additional particular stock, the fund can always gain a positive survival probability, regardless of parameter values of assets. Note that the setting in this paper potentially could be generalized to a model with objective to maximize expected utility. The impact of risk aversion and fixed cost on portfolio choice and consumption plan could be nontrivial. However, with assumptions on both utility function of consumption and fixed cost, it could be difficult to obtain closed-form solutions of optimal portfolio and optimal consumption. We leave it for future research.

## Appendix

### Derivation of Bellman equation for benchmark model

We now solve for the ruin probability using the martingale technique pioneer by Fleming and Rishel (1975). Define

$$M_t \equiv \int_{s=0}^t dt_s + P(w_t, t_t),$$

for  $t > 0$ , where  $P(\cdot, \cdot)$  is the proposed value function defined as  $1 - P^s$  and  $P^s$  is given in (8) in the statement of Proposition 1. This denotes the probability of subsequent termination in the future after date  $t$ . Note that by the dynamics of wealth (4) and by Itô's Lemma (letting  $dt^2 = dt dZ = 0$ ,  $dZ^2 = dt$ ), we have

$$\begin{aligned} dM_t &= dt_t + P_w dw_t + \frac{1}{2} P_{ww} (dw_t)^2 + (P(w_t, 1) - P(w_t, 0)) dt_t \\ &= dt_t + (1 - t_t) P_w (\mu w_t dt + \sigma w dZ_t - k_0 dt) + (P(w_t, 1) - P(w_t, 0)) dt_t. \end{aligned}$$

Moreover, the ruin probability as a conditional expectation should be a martingale. Because, a martingale do not have tendency to increase or decrease on average, and it well fit the role that it is bounded between 0 and 1 as a probability. Hence, the drift of  $M$  should be zero, gives rise to

$$0 = \frac{E[dM]}{dt} = \frac{E\left[(\mu w_t dt + \sigma w dZ_t - k_0 dt) P_w + \frac{1}{2} \sigma^2 w^2 P_{ww} dt\right]}{dt},$$

which yields the following ODE for ruin probability at any finite time,

$$(\mu w_t - k_0) P_w + \frac{1}{2} \sigma^2 w^2 P_{ww} = 0.$$

### Verifying solution and derivation of ruin probability

To verify

$$P = C_1 \int_0^w \exp\left(-\frac{2k_0}{\sigma^2 \varpi}\right) \varpi^{-\frac{2\mu}{\sigma^2}} d\varpi + C_2$$

is the solution of the ODE, we just need to derive

$$\begin{aligned} P_w &= C_1 \exp\left(-\frac{2k_0}{\sigma^2 w}\right) w^{-\frac{2\mu}{\sigma^2}}, \\ P_{ww} &= C_1 \left[ \frac{2k_0}{\sigma^2 w^2} \exp\left(-\frac{2k_0}{\sigma^2 w}\right) w^{-\frac{2\mu}{\sigma^2}} - \frac{2\mu}{\sigma^2} \exp\left(-\frac{2k_0}{\sigma^2 w}\right) w^{-\frac{2\mu}{\sigma^2}-1} \right], \end{aligned}$$

therefore,

$$\frac{P_{ww}}{P_w} = \frac{2(k_0 - \mu w_t)}{\sigma^2 w^2},$$

which is just another form of the ODE (5). This verifies the solution.

Now we derive the analytical expression of ruin probability when  $\mu/\sigma^2 = 1$ . We have

$$P = C_1 \int_0^w \exp\left(-\frac{2k_0}{\sigma^2 \varpi}\right) \varpi^{-2} d\varpi + C_2.$$

Note that  $-\varpi^{-2} d\varpi = d\varpi^{-1}$ , thus

$$\begin{aligned} P &= C_1 \frac{\sigma^2}{2k_0} \int_0^w \exp\left(-\frac{2k_0}{\sigma^2} \varpi^{-1}\right) d\left(-\frac{2k_0}{\sigma^2} \varpi^{-1}\right) + C_2 \\ &= C_1 \frac{\sigma^2}{2k_0} \exp\left(-\frac{2k_0}{\sigma^2 \varpi}\right) + C_2. \end{aligned}$$

Substitute the boundary conditions of

$$P(w = 0) = 1 \quad \text{and} \quad P(w = \infty) = 0,$$

we have

$$1 = C_2 + 0 \quad \text{and} \quad 0 = C_2 + \frac{\sigma^2}{2k_0} C_1,$$

hence, we obtain

$$C_1 = -\frac{2k_0}{\sigma^2} \quad \text{and} \quad C_2 = 1,$$

and therefore

$$P = 1 - \exp\left(-\frac{2k_0}{\sigma^2 w}\right).$$

□

## Proof of Proposition 2

### Derivation of ruin probability

We now derive the solution of the following Bellman equation which is an ODE,

$$\min_{\theta_t} \left[ (w(\mu + \theta_t \hat{\Delta}) - k_0) P_w + \frac{1}{2} (\sigma^2 + 2\theta \sigma (\sigma_1 - \sigma) + \theta_t^2 \hat{\sigma}^2) w^2 P_{ww} \right] = 0. \quad (21)$$

The first-order condition with respect to  $\theta$  gives

$$\hat{\Delta} w P_w + \sigma (\sigma_1 - \sigma) w^2 P_{ww} + \theta_t \hat{\sigma}^2 w^2 P_{ww} = 0 \implies \theta_t = -\frac{\hat{\Delta} P_w}{\hat{\sigma}^2 w P_{ww}} - \frac{\sigma (\sigma_1 - \sigma)}{\hat{\sigma}^2}.$$

Substitute it back to ODE (21) yields

$$awP_w - k_0 + bw^2 \frac{P_{ww}}{P_w} - \frac{\hat{\Delta}^2 P_w}{2\hat{\sigma}^2 P_{ww}} = 0$$

Let

$$g(w) \equiv \frac{P_{ww}}{P_w} = \frac{\partial \ln(-P_w)}{\partial w}, \quad (22)$$

we can rewrite the ODE as

$$aw - k_0 + bw^2 g - \frac{\hat{\Delta}^2}{2\hat{\sigma}^2} \frac{1}{g} = 0 \Leftrightarrow bw^2 g^2 + (aw - k_0)g - \frac{\hat{\Delta}^2}{2\hat{\sigma}^2} = 0.$$

Let

$$\bar{a} = bw^2, \bar{b} = aw - k_0, \bar{c} = -\frac{\hat{\Delta}^2}{2\hat{\sigma}^2},$$

we have

$$g_1(w) = \frac{-\bar{b} + \sqrt{\bar{b}^2 - 4\bar{a}\bar{c}}}{2\bar{a}} = \frac{k_0 - aw + \sqrt{(aw - k_0)^2 + \frac{2bw^2\hat{\Delta}^2}{\hat{\sigma}^2}}}{2bw^2} > 0,$$

$$\text{and } g_2(w) = \frac{-\bar{b} - \sqrt{\bar{b}^2 - 4\bar{a}\bar{c}}}{2\bar{a}} = \frac{k_0 - aw - \sqrt{(aw - k_0)^2 + \frac{2bw^2\hat{\Delta}^2}{\hat{\sigma}^2}}}{2bw^2} < 0.$$

Note that the ruin probability decreases with wealth, therefore, we have  $P_w < 0$ . Since we need to assume that  $P_{ww} > 0$  to ensure a minimization problem, by the expression of optimal portfolio, we have  $g = P_{ww}/P_w < 0$ , therefore, we have  $g = g_2(w)$ . Now to obtain the ruin probability, we only need to solve Eq. (22) with  $g = g_2(w)$ .

Integration of both side of the Eq. (22) form 1 to  $w$  yields

$$\int_1^w g(\varpi) d\varpi = \ln(-P_w) + \bar{C}_1.$$

Integrating again from 0 to  $w$  yields the general solution of the ruin probability,

$$P(w) = \hat{C}_1 \int_0^w \exp\left(\int_1^h g(\varpi) d\varpi\right) dh + \hat{C}_2, \quad (23)$$

where constant  $\hat{C}_1$  and  $\hat{C}_2$  is to be determined by boundary conditions. Similar to the discussion in the previous section, we also have the two boundary conditions as

$$P(w=0) = 1 \quad \text{and} \quad P(w=\infty) = 0.$$

Imposing the two boundary conditions to (23) gives

$$\hat{C}_1 = -\frac{1}{\int_0^\infty \left[ \exp\left(\int_1^h g(\varpi) d\varpi\right) \right] dh} \quad \text{and} \quad \hat{C}_2 = 1.$$

Hence, to calculate  $\hat{C}_1$  and, thus, the general solution, we need to first solve for function  $\int_1^h g(\varpi) d\varpi$ . The following subsection provides an analytical expression of the indefinite integral  $\int g(\varpi) d\varpi$ .

### Calculate integral

By the expression of  $g_2(\varpi)$ , we have

$$\begin{aligned}\int g_2(\varpi) d\varpi &= \int \frac{k_0 - a\varpi - \sqrt{(a\varpi - k_0)^2 + 2b\varpi^2\hat{\Delta}^2/\hat{\sigma}^2}}{2b\varpi^2} d\varpi \\ &= \frac{1}{2b} \int \frac{k_0 - a\varpi}{\varpi^2} d\varpi - \frac{1}{2b} \int \frac{\sqrt{(a\varpi - k_0)^2 + 2b\varpi^2\hat{\Delta}^2/\hat{\sigma}^2}}{\varpi^2} d\varpi.\end{aligned}$$

Rearranging the first integral, we have

$$\int \frac{k_0 - a\varpi}{\varpi^2} d\varpi = k_0 \int \frac{1}{\varpi^2} d\varpi - a \int \frac{1}{\varpi} d\varpi = -k_0 \frac{1}{\varpi} - a \ln \varpi.$$

Rearranging the second integral, and let  $\bar{\mu}^2 = a^2 + 2b\hat{\Delta}^2/\hat{\sigma}^2$ , we have

$$SI \equiv \int \frac{\sqrt{(a\varpi - k_0)^2 + 2b\varpi^2\hat{\Delta}^2/\hat{\sigma}^2}}{\varpi^2} d\varpi = \int \frac{1}{\varpi} \sqrt{\bar{\mu}^2 - \frac{2ak_0}{\varpi} + \frac{k_0^2}{\varpi^2}} d\varpi.$$

Let  $t = 1/\varpi \Rightarrow 1/t = \varpi$  and  $-(1/t^2)dt = d\varpi$ , hence the integral becomes

$$SI = \int \frac{1}{\varpi} \sqrt{\bar{\mu}^2 - \frac{2ak_0}{\varpi} + \frac{k_0^2}{\varpi^2}} d\varpi = - \int \frac{1}{t} \sqrt{k_0^2 t^2 - 2ak_0 t + \bar{\mu}^2} dt.$$

Let

$$\sqrt{k_0^2 t^2 - 2ak_0 t + \bar{\mu}^2} = k_0 t - x,$$

we have

$$\begin{aligned}k_0^2 t^2 - 2ak_0 t + \bar{\mu}^2 &= k_0^2 t^2 + x^2 - 2k_0 t x \\ \Rightarrow 2k_0 t x - 2ak_0 t &= x^2 - \bar{\mu}^2 \quad \text{and} \quad t = \frac{x^2 - \bar{\mu}^2}{2k_0 x - 2k_0 a}.\end{aligned}$$

Note that

$$\left( \frac{x^2 - \bar{\mu}^2}{2k_0(x - a)} \right)' = - \frac{(x^2 - \bar{\mu}^2)}{2k_0(x - a)^2} + \frac{2x(x - a)}{2k_0(x - a)^2} = \frac{x^2 - 2xa + \bar{\mu}^2}{2k_0(x - a)^2},$$

therefore

$$dt = \frac{x^2 - 2xa + \bar{\mu}^2}{2k_0(x-a)^2} dx.$$

Moreover, we have

$$\begin{aligned} \sqrt{k_0^2 t^2 - 2ak_0 t + \bar{\mu}^2} &= k_0 t - x \\ &= \frac{k_0 x^2 - k_0 \bar{\mu}^2}{2k_0 x - 2k_0 a} - \frac{2k_0 x^2 - 2k_0 ax}{2k_0 x - 2k_0 a} = \frac{-k_0 \bar{\mu}^2 - k_0 x^2 + 2k_0 ax}{2k_0 x - 2k_0 a}. \end{aligned}$$

Therefore,

$$\begin{aligned} SI &= - \int \frac{1}{t} \sqrt{k_0^2 t^2 - 2ak_0 t + \bar{\mu}^2} dt \\ &= - \int \frac{2k_0 x - 2k_0 a}{x^2 - \bar{\mu}^2} \frac{-k_0 \bar{\mu}^2 - k_0 x^2 + 2k_0 ax}{2k_0 x - 2k_0 a} \frac{x^2 - 2xa + \bar{\mu}^2}{2k_0(x-a)^2} dx \\ &= \frac{1}{2} \int \frac{(\bar{\mu}^2 + x^2 - 2ax)^2}{(x-a)^2(x^2 - \bar{\mu}^2)} dx. \end{aligned}$$

We can express the above integrand as a sum of easily integrable functions

$$\frac{(\bar{\mu}^2 + x^2 - 2ax)^2}{(x-a)^2(x^2 - \bar{\mu}^2)} = \frac{x}{x-a} + \frac{A}{x-a} + \frac{B}{(x-a)^2} + \frac{C}{x - \sqrt{\bar{\mu}^2}} + \frac{D}{x + \sqrt{\bar{\mu}^2}},$$

where constants  $A, B, C$ , and  $D$  are to determined. Thus, we have

$$\begin{aligned} (\bar{\mu}^2 + x^2 - 2ax)^2 &= x(x-a)(x^2 - \bar{\mu}^2) + A(x-a)(x^2 - \bar{\mu}^2) + B(x^2 - \bar{\mu}^2) \\ &\quad + C(x-a)^2(x + \sqrt{\bar{\mu}^2}) + D(x-a)^2(x - \sqrt{\bar{\mu}^2}). \end{aligned}$$

Let  $x = a$ , we have

$$(\bar{\mu}^2 + a^2 - 2a^2)^2 = B(a^2 - \bar{\mu}^2) \implies B = a^2 - \bar{\mu}^2.$$

Let  $x = \sqrt{\bar{\mu}^2}$ , we have

$$\begin{aligned} (\bar{\mu}^2 + x^2 - 2ax)^2 &= C(x-a)^2(x + \sqrt{\bar{\mu}^2}) \iff 2(\bar{\mu}^2 - a\sqrt{\bar{\mu}^2})^2 = C\sqrt{\bar{\mu}^2}(\sqrt{\bar{\mu}^2} - a)^2 \\ &\implies C = 2\sqrt{\bar{\mu}^2}. \end{aligned}$$

Let  $x = -\sqrt{\bar{\mu}^2}$ , we have

$$\begin{aligned} (\bar{\mu}^2 + x^2 - 2ax)^2 &= D(x-a)^2(x - \sqrt{\bar{\mu}^2}) \\ &\implies D = -\frac{(\bar{\mu}^2 + \bar{\mu}^2 + 2a\sqrt{\bar{\mu}^2})^2}{(\sqrt{\bar{\mu}^2} + a)^2 2\sqrt{\bar{\mu}^2}} = -2\sqrt{\bar{\mu}^2}. \end{aligned}$$

Let  $x = 0$ , we have

$$\begin{aligned}\bar{\mu}^4 &= Aa\bar{\mu}^2 - B\bar{\mu}^2 + Ca^2\sqrt{\bar{\mu}^2} - Da^2\sqrt{\bar{\mu}^2} = Aa\bar{\mu}^2 - (a^2 - \bar{\mu}^2)\bar{\mu}^2 + 4\bar{\mu}^2a^2 \\ \implies \bar{\mu}^2 &= Aa - (a^2 - \bar{\mu}^2) + 4a^2 = A + 3a \implies A = -3a.\end{aligned}$$

Thus, we have

$$\begin{aligned}&\int \frac{(\bar{\mu}^2 + x^2 - 2ax)^2}{(x-a)^2(x^2 - \bar{\mu}^2)} dx \\ &= \int \left( \frac{x}{x-a} + \frac{A}{x-a} + \frac{B}{(x-a)^2} + \frac{C}{x-\sqrt{\bar{\mu}^2}} + \frac{D}{x+\sqrt{\bar{\mu}^2}} \right) dx \\ &= x + (a+A) \ln |x-a| - B \frac{1}{x-a} + C \ln |x-\bar{\mu}| + D \ln |x+\bar{\mu}|.\end{aligned}$$

Therefore, eventually, we have

$$\begin{aligned}\int g(\varpi) d\varpi &= \frac{1}{2b} \int \frac{k_0 - a\varpi}{\varpi^2} d\varpi - \frac{1}{2b} \int \frac{\sqrt{(a\varpi - k_0)^2 + 2b\varpi^2 \hat{\Delta}^2 / \hat{\sigma}^2}}{\varpi^2} d\varpi \\ &= \frac{1}{2b} \left( -k_0 \frac{1}{\varpi} - a \ln \varpi \right) - \frac{1}{4b} \left( x + (a+A) \ln |a-x| - B \frac{1}{x-a} \right. \\ &\quad \left. + C \ln |\bar{\mu}-x| + D \ln |x+\bar{\mu}| \right),\end{aligned}$$

where

$$x = k_0 t - \sqrt{k_0^2 t^2 - 2ak_0 t + \bar{\mu}^2} = \frac{k_0}{\varpi} - \sqrt{\frac{k_0^2}{\varpi^2} - \frac{2ak_0}{\varpi} + \bar{\mu}^2}.$$

### Simplified expression for ruin probability

With the expression of  $\int g(\varpi) d\varpi$ , we have

$$\begin{aligned}\exp \left( \int g(\varpi) d\varpi \right) &= \exp \left( -\frac{k_0}{2b} \frac{1}{\varpi} + \ln \varpi^{-\frac{a}{2b}} \right) \\ &\quad \times \exp \left( -\frac{x}{4b} + \ln |a-x|^{-\frac{a+A}{4b}} + \frac{B}{4b} \frac{1}{(x-a)} + \ln |\bar{\mu}-x|^{-\frac{C}{4b}} + \ln |x+\bar{\mu}|^{-\frac{D}{4b}} \right) \\ &= \varpi^{-\frac{a}{2b}} (|a-x|)^{-\frac{a+A}{4b}} \left( \left| \frac{x+\bar{\mu}}{\bar{\mu}-x} \right| \right)^{\frac{C}{4b}} \exp \left( \frac{B}{4b} \frac{1}{(x-a)} - \frac{k_0}{2b} \frac{1}{\varpi} - \frac{x}{4b} \right) \\ &= \left( \left| \frac{a-x}{\varpi} \right| \right)^{\frac{a}{2b}} \left( \left| \frac{\bar{\mu}+x}{\bar{\mu}-x} \right| \right)^{\frac{\bar{\mu}}{2b}} \exp \left( \frac{a^2 - \bar{\mu}^2}{4b(x-a)} - \frac{k_0}{2b\varpi} - \frac{x}{4b} \right).\end{aligned}$$

Note that in a form of definite integral, we have

$$\begin{aligned}\exp \left( \int_1^h g(\varpi) d\varpi \right) &= \exp \left( \int g(\varpi) d\varpi \Big|_{\varpi=h} - \int g(\varpi) d\varpi \Big|_{\varpi=1} \right) \\ &= \exp \left( - \int g(\varpi) d\varpi \Big|_{\varpi=1} \right) \exp \left( \int g(\varpi) d\varpi \Big|_{\varpi=h} \right),\end{aligned}$$

therefore,

$$\int_0^w \exp \left( \int_1^h g(\varpi) ds \right) dh = \exp \left( - \int g(\varpi) d\varpi \Big|_{\varpi=1} \right) \int_0^w \exp \left( \int g(\varpi) d\varpi \Big|_{\varpi=h} \right) dh.$$

Substitute it back to (23), we obtain the expression for ruin probability.

### Derivation of optimal portfolio weight

With the expression of optimal portfolio weight in asset  $B$ , when  $P_{ww} > 0$ , we have

$$\begin{aligned} \theta &= -\frac{\hat{\Delta} P_w}{\hat{\sigma}^2 w P_{ww}} - \frac{\sigma(\sigma_1 - \sigma)}{\hat{\sigma}^2} \\ &= -\frac{\hat{\Delta}}{\hat{\sigma}^2 w} \frac{2bw^2}{k_0 - aw - \sqrt{(aw - k_0)^2 + 2bw^2 \hat{\Delta}^2 / \hat{\sigma}^2}} - \frac{\sigma(\sigma_1 - \sigma)}{\hat{\sigma}^2} \\ &= \frac{1}{\hat{\Delta}} \left( \frac{k_0}{w} - a + \sqrt{\left( a - \frac{k_0}{w} \right)^2 + \frac{2b\hat{\Delta}^2}{\hat{\sigma}^2}} \right) - \frac{\sigma(\sigma_1 - \sigma)}{\hat{\sigma}^2}. \end{aligned}$$

Moreover, the absolute portfolio weight in asset  $B$  is given as  $\theta w$ , hence, we have  $\lim_{w \rightarrow 0} \theta w = 2k_0 / \hat{\Delta}$ .

### Lemma 2 and Proof

**Lemma 5** Given  $x = k_0 / \varpi - \sqrt{k_0^2 / \varpi^2 - 2ak_0 / \varpi + \bar{\mu}^2}$ , and  $\bar{\mu}^2 = a^2 + 2b\hat{\Delta}^2 / \hat{\sigma}^2$ , we have  $(\bar{\mu} + x) / (\bar{\mu} - x) \sim 1 / \varpi$ .

**Proof** We now prove  $(\bar{\mu} + x) / (\bar{\mu} - x) \sim 1 / \varpi$  as  $\varpi \rightarrow \infty$  and  $x - a \sim \varpi$  as  $\varpi \rightarrow 0$ . Since  $\lim_{\varpi \rightarrow \infty} x = -\bar{\mu}$ , hence  $\lim_{\varpi \rightarrow \infty} \bar{\mu} - x = -2\bar{\mu}$ . Therefore, to proof  $\frac{\bar{\mu}+x}{\bar{\mu}-x} \sim \frac{1}{\varpi}$ , we only need to proof  $\bar{\mu} + x \sim \frac{1}{\varpi}$ , which is easy to see by

$$\bar{\mu} + x = \bar{\mu} + \frac{k_0}{\varpi} - \sqrt{\frac{k_0^2}{\varpi^2} - \frac{2ak_0}{\varpi} + \bar{\mu}^2} = \frac{1}{\varpi} \frac{2k_0(\bar{\mu} + a)}{\bar{\mu} + \frac{k_0}{\varpi} + \sqrt{\frac{k_0^2}{\varpi^2} - \frac{2ak_0}{\varpi} + \bar{\mu}^2}},$$

where

$$\lim_{\varpi \rightarrow \infty} \frac{2k_0(\bar{\mu} + a)}{\bar{\mu} + \frac{k_0}{\varpi} + \sqrt{\frac{k_0^2}{\varpi^2} - \frac{2ak_0}{\varpi} + \bar{\mu}^2}} = \frac{k_0(\bar{\mu} + a)}{\bar{\mu}}.$$

Moreover, note that

$$\begin{aligned}
 x - a &= \frac{\left(\frac{k_0}{\varpi} - a - \sqrt{\frac{k_0^2}{\varpi^2} - \frac{2ak_0}{\varpi} + \bar{\mu}^2}\right) \left(\frac{k_0}{\varpi} - a + \sqrt{\frac{k_0^2}{\varpi^2} - \frac{2ak_0}{\varpi} + \bar{\mu}^2}\right)}{\frac{k_0}{\varpi} - a + \sqrt{\frac{k_0^2}{\varpi^2} - \frac{2ak_0}{\varpi} + \bar{\mu}^2}} \\
 &= \varpi \frac{a^2 - \bar{\mu}^2}{k_0 - \varpi a + \sqrt{k_0^2 - 2ak_0\varpi + \bar{\mu}^2\varpi^2}}.
 \end{aligned}$$

Hence, we have

$$\lim_{\varpi \rightarrow 0} \frac{x - a}{\varpi} = \frac{a^2 - \bar{\mu}^2}{k_0 + k_0} = \frac{a^2 - \bar{\mu}^2}{2k_0},$$

and this completes the proof.  $\square$

### Derivations of optimal portfolio and Bellman equation

By the dynamics of wealth (10) and by Itô's Lemma (letting  $dt^2 = dt dZ = 0$ ,  $dZ^2 = dt$ ), we have

$$\begin{aligned}
 dM_t &= dI_t + P_w dw_t + \frac{1}{2} P_{ww} (dw_t)^2 + (P(w_t, 1) - P(w_t, 0)) dI_t \\
 &= dI_t + (P(w_t, 1) - P(w_t, 0)) dI_t, \\
 &\quad + (1 - I_t) P_w (w(\mu dt + \sigma dZ_t + \theta_t((\mu_1 + \Delta - \mu)dt + (\sigma_1 - \sigma)dZ_t + \sigma_0 dZ_t^0)) - k_0 dt) \\
 &\quad + \frac{1}{2} (1 - I_t)^2 P_{ww} w^2 (\sigma^2 + 2\theta_t \sigma(\sigma_1 - \sigma) + \theta_t^2 ((\sigma_1 - \sigma)^2 + \sigma_0^2)) dt
 \end{aligned}$$

where we employ

$$\begin{aligned}
 (dw)^2 &= (w(\mu dt + \sigma dZ_t + \theta_t((\mu_1 + \Delta - \mu)dt + (\sigma_1 - \sigma)dZ_t + \sigma_0 dZ_t^0)) - k_0 dt)^2 \\
 &= w^2 (\theta_t^2 ((\sigma_1 - \sigma)^2 + \sigma_0^2) + 2\theta_t(\sigma_1 - \sigma)\sigma + \sigma^2) dt.
 \end{aligned}$$

For the optimal portfolio,  $M$  should be a martingale, hence, the drift of  $M$  should be zero, which yields the following Bellman equation as an ODE

$$\min_{\theta_t} \left[ (w(\mu + \theta_t \hat{\Delta}) - k_0) P_w + \frac{1}{2} (\sigma^2 + 2\theta_t \sigma(\sigma_1 - \sigma) + \theta_t^2 \hat{\sigma}^2) w^2 P_{ww} \right] = 0.$$

where  $\hat{\Delta} = \mu_1 - \mu + \Delta$ , and  $\hat{\sigma}^2 = \sigma_0^2 + (\sigma_1 - \sigma)^2$ . Rearranging, we have

$$\min_{\theta_t} \left[ \mu w P_w + \theta_t \hat{\Delta} w P_w - k_0 P_w + \frac{1}{2} \sigma^2 w^2 P_{ww} + \theta_t \sigma(\sigma_1 - \sigma) w^2 P_{ww} + \frac{1}{2} \theta_t^2 \hat{\sigma}^2 w^2 P_{ww} = 0 \right]. \quad (24)$$

First-order condition w.r.t.  $\theta_t$  gives

$$\hat{\Delta} w P_w + \sigma(\sigma_1 - \sigma) w^2 P_{ww} + \theta_t \hat{\sigma}^2 w^2 P_{ww} = 0.$$

Thus, we obtain the optimal portfolio as

$$\theta_t^* = -\frac{\hat{\Delta}P_w + \sigma(\sigma_1 - \sigma)wP_{ww}}{\hat{\sigma}^2wP_{ww}} = -\frac{\hat{\Delta}P_w}{\hat{\sigma}^2wP_{ww}} + \phi,$$

where  $\phi = -\frac{\sigma(\sigma_1 - \sigma)}{\hat{\sigma}^2}$ . Substitute the optimal portfolio back to the ODE (24), we have

$$\begin{aligned} & \left( w \left( \mu + \left( -\frac{\hat{\Delta}P_w}{\hat{\sigma}^2wP_{ww}} + \phi \right) \hat{\Delta} \right) - k_0 \right) P_w \\ & + \frac{1}{2} \left( \sigma^2 + 2 \left( -\frac{\hat{\Delta}P_w}{\hat{\sigma}^2wP_{ww}} + \phi \right) \sigma(\sigma_1 - \sigma) + \left( -\frac{\hat{\Delta}P_w}{\hat{\sigma}^2wP_{ww}} + \phi \right)^2 \hat{\sigma}^2 \right) w^2 P_{ww} = 0 \\ \Leftrightarrow & \mu w P_w + \phi \hat{\Delta} w P_w - \phi \hat{\Delta} w P_w - \frac{\hat{\Delta} \sigma (\sigma_1 - \sigma)}{\hat{\sigma}^2} w P_w + \frac{\hat{\Delta}^2 P_w^2}{2 \hat{\sigma}^2 P_{ww}} - \frac{\hat{\Delta}^2 P_w^2}{\hat{\sigma}^2 P_{ww}} - k_0 P_w \\ & + \phi \sigma (\sigma_1 - \sigma) w^2 P_{ww} + \frac{\sigma^2 w^2 P_{ww}}{2} + \frac{\phi^2 \hat{\sigma}^2 w^2 P_{ww}}{2} = 0 \\ \Leftrightarrow & \left( \mu - \frac{\hat{\Delta} \sigma (\sigma_1 - \sigma)}{\hat{\sigma}^2} \right) w P_w - k_0 P_w - \frac{\hat{\Delta}^2 P_w^2}{2 \hat{\sigma}^2 P_{ww}} \\ & + \left( \phi \sigma (\sigma_1 - \sigma) + \frac{\sigma^2 + \phi^2 \hat{\sigma}^2}{2} \right) w^2 P_{ww} = 0 \\ \Leftrightarrow & a w P_w - k_0 P_w - \frac{\hat{\Delta}^2 P_w^2}{2 \hat{\sigma}^2 P_{ww}} + b w^2 P_{ww} = 0, \end{aligned}$$

where

$$a \equiv \mu - \frac{\hat{\Delta} \varphi}{\hat{\sigma}^2}, \quad b \equiv \frac{\sigma^2 \sigma_0^2}{2 \hat{\sigma}^2}, \quad \phi \equiv -\frac{\varphi}{\hat{\sigma}^2}, \quad \text{and } \varphi \equiv \sigma(\sigma_1 - \sigma).$$

Note that

$$b = \phi \sigma (\sigma_1 - \sigma) + \frac{\sigma^2 + \phi^2 \hat{\sigma}^2}{2} = \frac{\sigma^2 \sigma_0^2}{2(\sigma_0^2 + (\sigma_1 - \sigma)^2)} = \frac{\sigma^2 \sigma_0^2}{2 \hat{\sigma}^2}.$$

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